

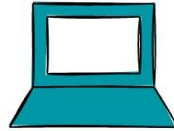
Quarkus demo - architectures outline :

Model enhancements types/architectures

(to be discussed during the Quarkus AI demo)



1: Query



*"Why is Camille's cat,
Mitsou, the cutest cat in
the whole world?"*

3: Response generation
Given:
+ query
+ relevant data/documents

(External) LLM
E.g., Mistral



4: Response output
response
+ potential citations
+ potential references

2: Return relevant
data/documents

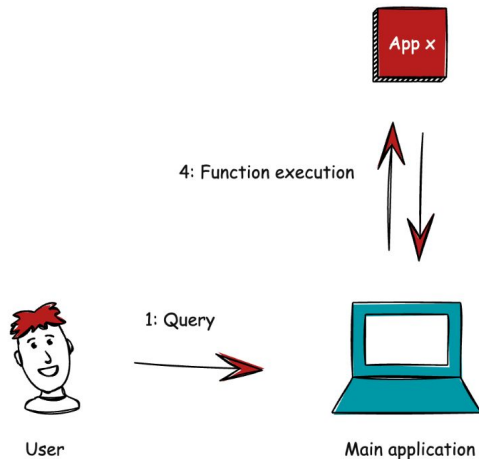
2: Search relevant
data/documents



Knowledge base

RAG

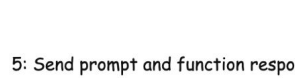
- https://redhat-developer-demos.github.io/quarkus-tutorial/quarkus-tutorial/17_prompts.html
- Protect data.
- No knowledge, use case, time, ... to fine-tune, train a model yourself.
- Source references.
- Less scalable, bigger models.
- Quarkus: next-gen java



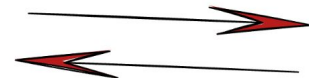
2: Send prompt and available functions



3: If response requires function call,
the LLM responds with function definition



5: Send prompt and function response



6: Final response is returned

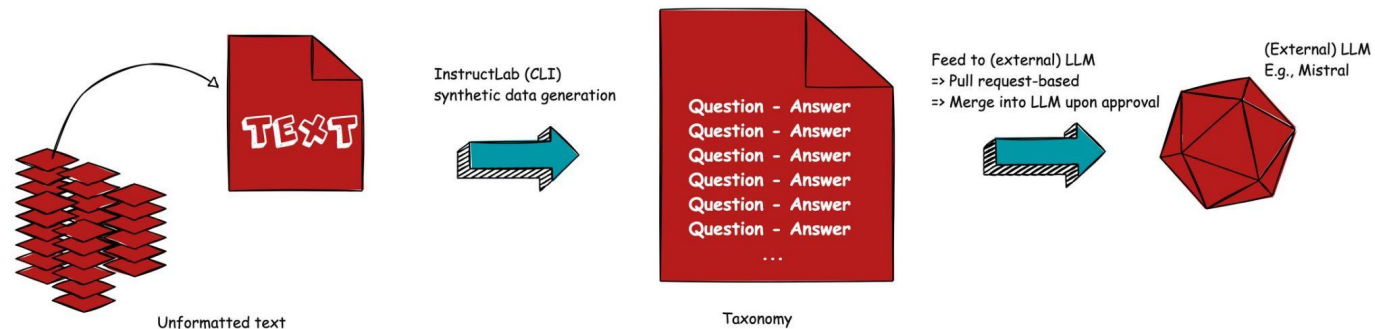
(External) LLM
E.g., Mistral



Function calling

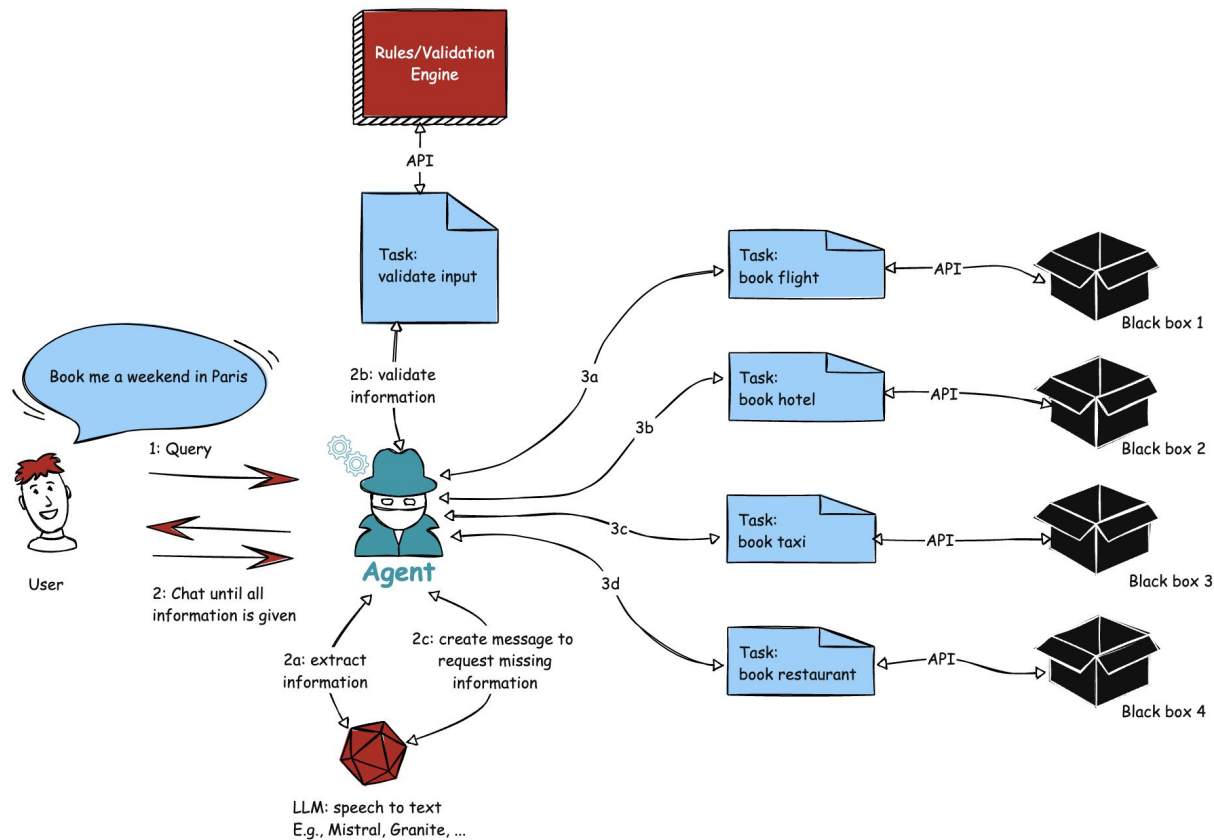
- https://redhat-developer-demos.github.io/quarkus-tutorial/quarkus-tutorial/17_prompts.html
- External call(-out).
- External APIs.
- Get factual data.

What is the weather today?



InstructLab

- <https://github.com/instructlab>
- Open collaboration.
- Open way of contributing.
- Less data science knowledge.



Agentic AI

- Quarkus.
- Camel.
- Kafka.
- ...
- Intern in prod!!



Embedding AI

Don't burn your fingers in the heat of
the AI revolution





Embedding AI

Staying cool behind the bar of the AI
revolution



**“Keep your
options open!”**

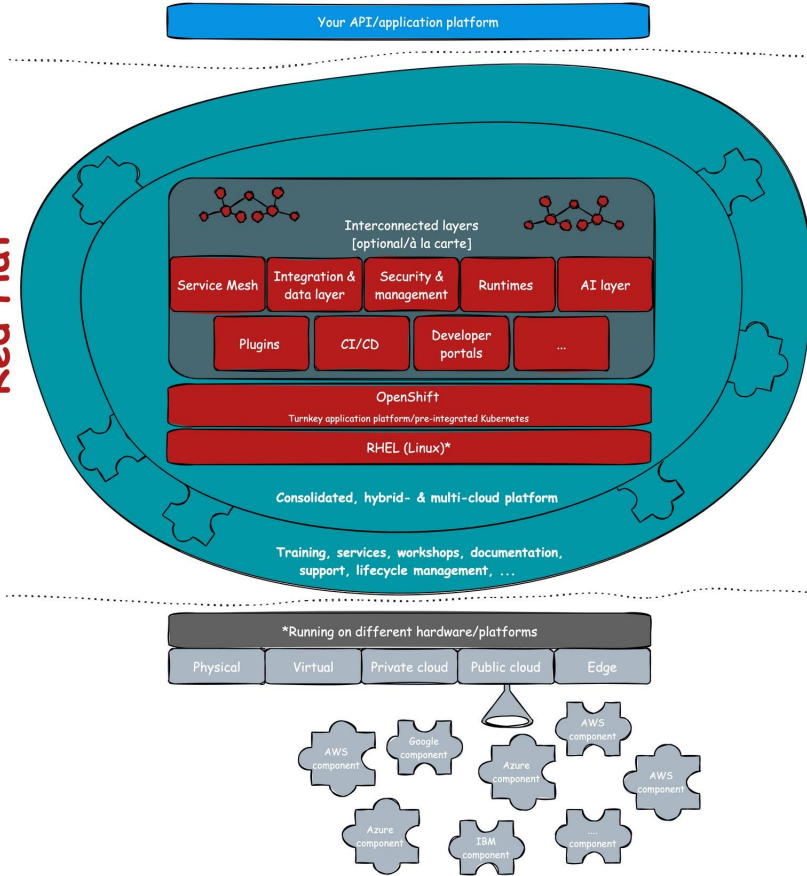


Maarten Vandepierre

maarten-vandepierre.be



Red Hat



Applications

App & AI platform
⇒ software factory

Hybrid infrastructure

**“Doing AI is more
than accessing a
model”**



A man with curly brown hair and glasses, wearing a dark jacket over a light-colored shirt, is looking down at a smartphone in his hands. He has a brown backpack strap over his shoulder. In the bottom left corner, there is a stack of papers and a dark mug. A large, white, cloud-shaped thought bubble with a thick black outline is positioned in the center of the image. It has a small tail pointing towards the man's head. The background is a vibrant red with numerous yellow and orange diagonal streaks radiating from behind the thought bubble, creating a sense of energy and focus.

Behind the bar of AI

What's really being served?

Real world news

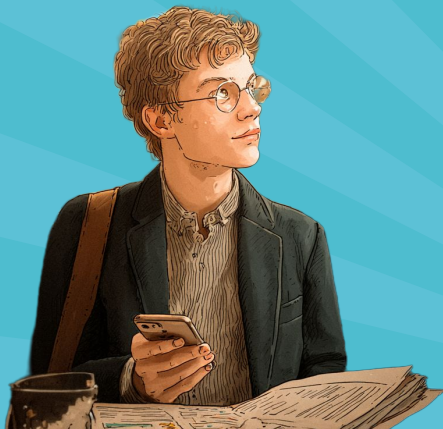
Cyber Security

What the CLOUD Act Really Means for EU Data Sovereignty

The CLOUD Act allows U.S. authorities to access data stored in the EU, putting it in direct conflict with GDPR. Learn how this impacts data sovereignty and what EU businesses can do to stay compliant

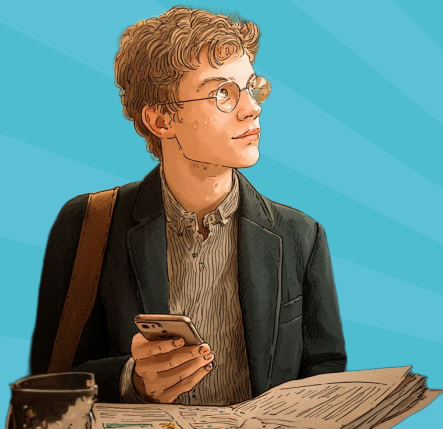


Wire
Jul 15, 2025



Source: <https://wire.com/en/blog/cloud-act-eu-data-sovereignty>

Real world news



EU data not protected from US access

21 July, 2025 11:35

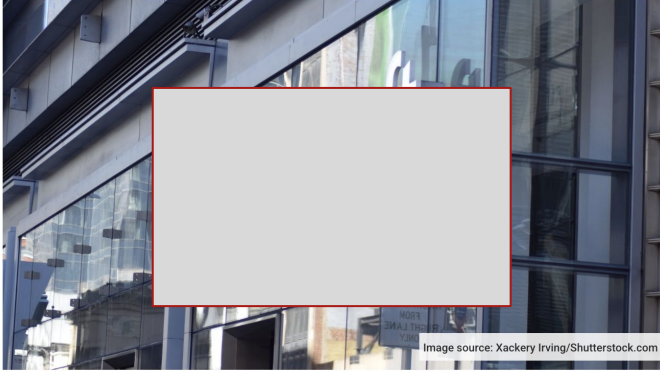


Image source: Xackery Irving/Shutterstock.com

Facebook

X

LinkedIn

Reddit

WhatsApp

Pocket

 This article is also available in German.

Explosive testimony before the French Senate: cannot guarantee that European user data is safe from US authorities despite EU hosting.

cannot protect European user data from access by US authorities – even if it is stored in EU data centers. This was admitted under oath by the legal director of France, , at a [hearing before the French Senate](#) on June 10, 2025.

Source: [https://www.it-daily.net/en/shortnews-en--eu-data-not-protected-from-us-access](https://www.it-daily.net/en/shortnews-en-<div></div>-eu-data-not-protected-from-us-access)

Real world news

Daryna Antoniuk

June 13th, 2025

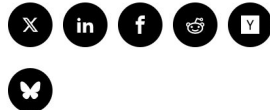
Technology

Government

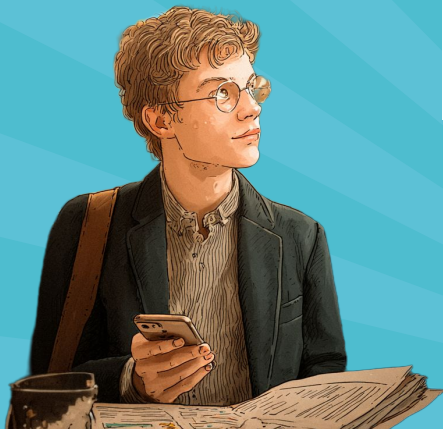
News

Danish government agency to ditch Microsoft software in push for digital independence

Denmark's tech modernization agency plans to replace Microsoft products with open-source software to reduce dependence on U.S. tech firms.



In an interview with the local newspaper *Politiken*, Danish Minister for Digitalisation Caroline Stage Olsen confirmed that over half of the ministry's staff will switch from Microsoft Office to LibreOffice next month, with a full transition to open-source software by the end of the year.



Source: <https://therecord.media/denmark-digital-agency-microsoft-digital-independence>

Real world news

NEWSLETTERS · CFO DAILY

MIT report: 95% of generative AI pilots at companies are failing



BY SHERYL ESTRADA

SENIOR WRITER AND AUTHOR OF CFO DAILY

August 18, 2025 at 6:54 AM EDT



GETTY IMAGES

Forbes

EDITORS' PICK | BUSINESS > MANUFACTURING

Why 95% Of AI Pilots Fail, And What Business Leaders Should Do Instead

By [Andrea Hill](#), Contributor. © Andrea Hill is a multi-industry CEO covering bu...

Follow Author

Published Aug 21, 2025, 11:54am EDT, Updated Aug 28, 2025, 07:54am EDT

Source: <https://www.forbes.com/sites/andreahill/2025/08/21/why-95-of-ai-pilots-fail-and-what-business-leaders-should-do-instead/>
<https://fortune.com/2025/08/18/mit-report-95-percent-generative-ai-pilots-at-companies-failing-cfo/>





Before we start pouring

Explaining key concepts



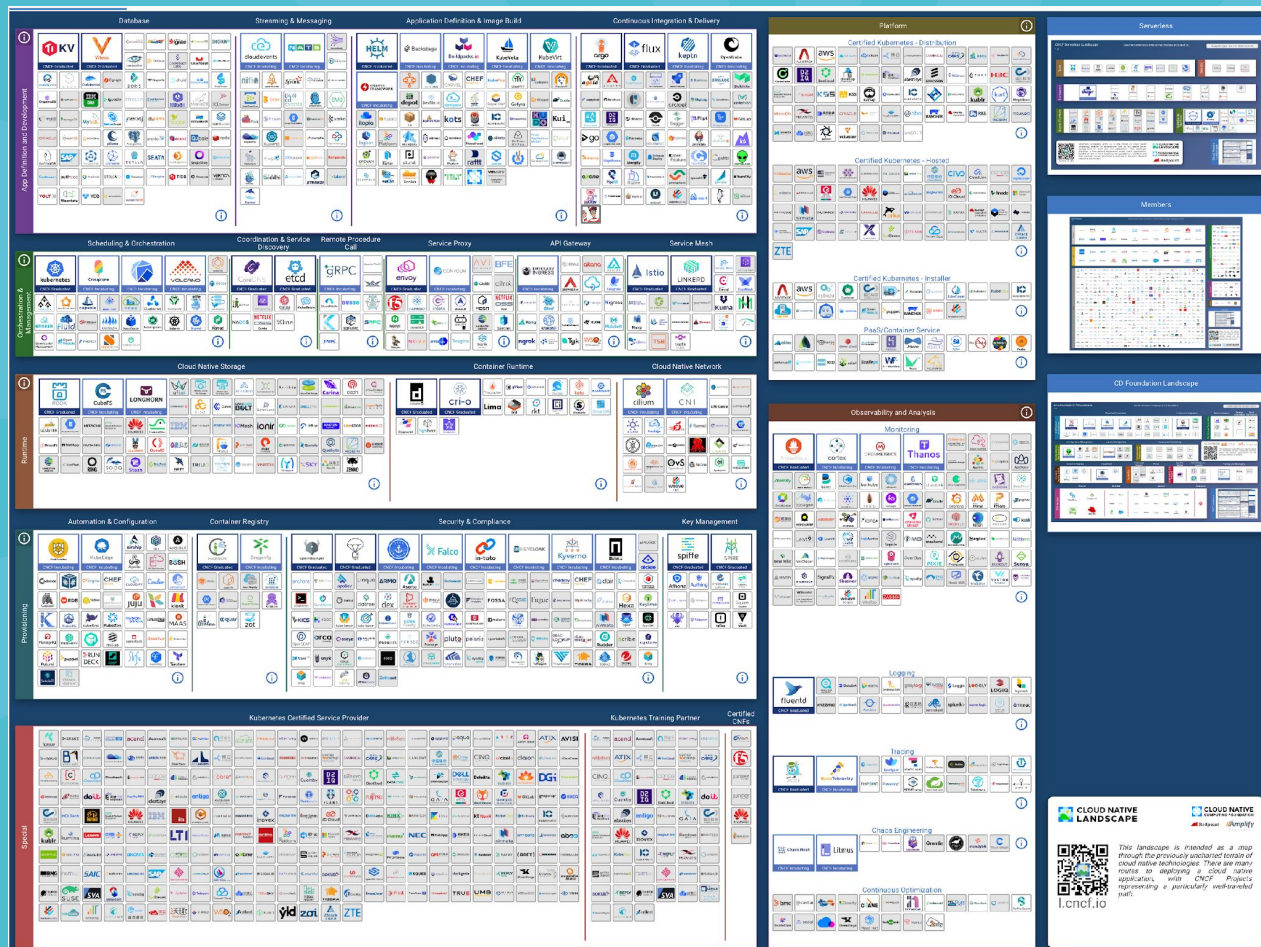
Cognitive load

The Cloud-Native Grocery Store

Where is

- Wally?
- Waldo?
- Charly?

Source: <https://landscape.cncf.io/>



Cognitive load

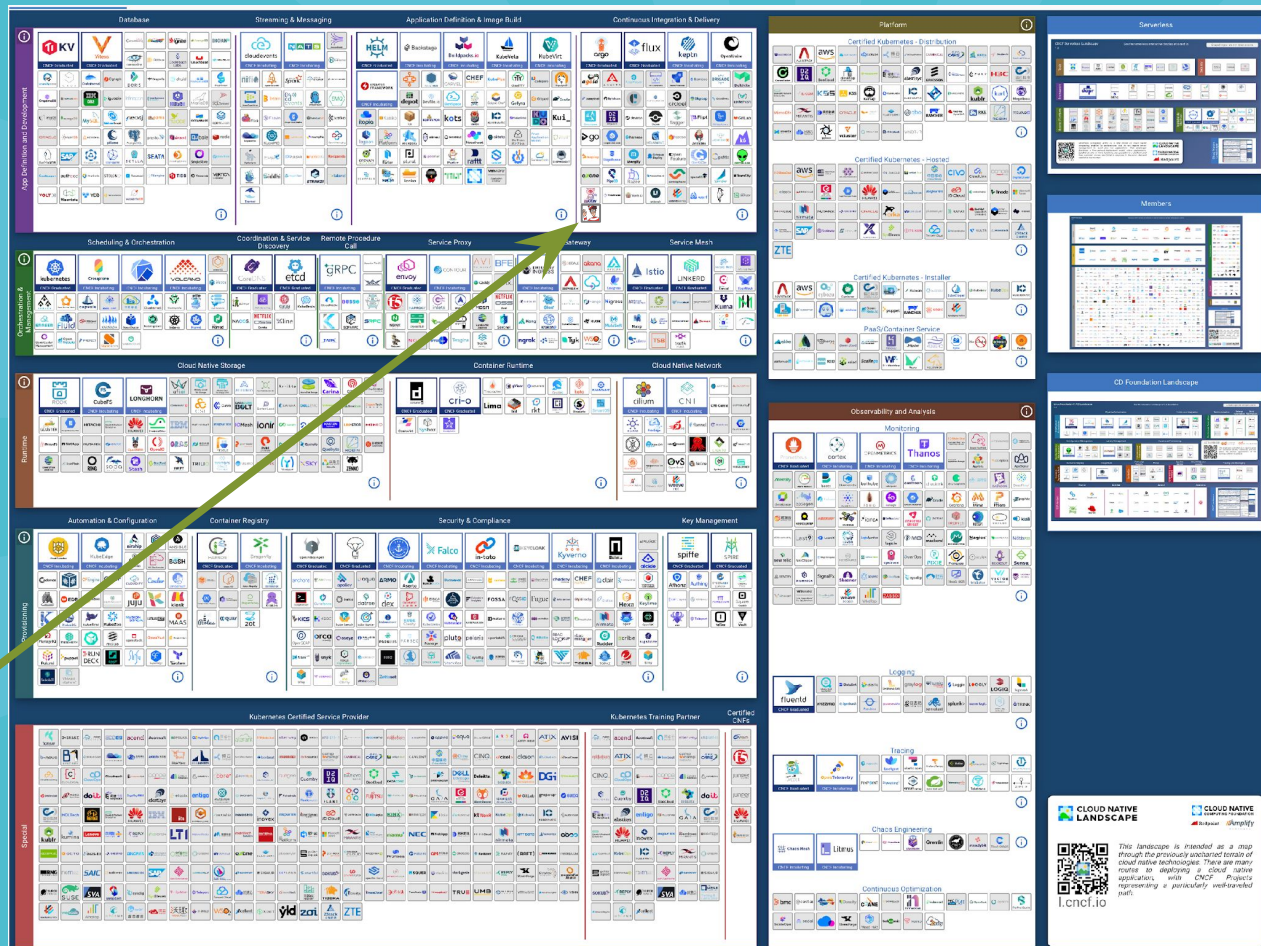
The Cloud-Native Grocery Store

Where is

- Wally?
- Waldo?
- Charly?



Source: <https://landscape.cncf.io/>



Platform engineering

Traditional App Delivery



Dev



Ops

Key:

Dev Responsible

Ops Responsible

Complete platform engineering presentation: https://youtube.com/watch?v=DaYNwrvJGmk&si=_IDX7xiBvBixEHPP

Platform engineering

Traditional App Delivery



Dev

Applications
App Deploy
App Build
Entitlement/Billing
Runtime
Java
Guest OS
Guest Virtualization
Clustering/HA
Host Virtualization
OS
Hardware



Ops



Key:

Dev Responsible

Ops Responsible

Modern App Delivery



Dev

Applications
App Deploy
App Build
Entitlement/Billing
Runtime
Java
Guest OS
Guest Virtualization
Clustering/HA
Host Virtualization
OS
Hardware



Ops

Usability

Observability

Quality

Platform engineering



Dev

Applications



Platform Engineering

App Build

App Deploy

Entitlement/Billing

Runtime

Java

Guest OS



Ops

Entitlement/Billing

Guest Virtualization

Clustering/HA

Host Virtualization

OS

Hardware

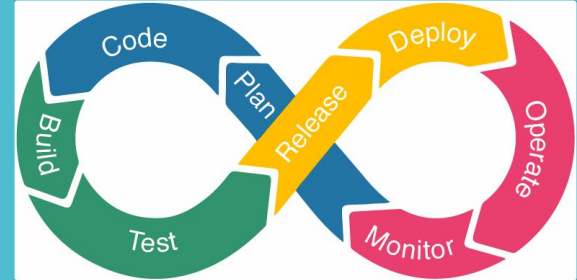
Internal Developer
Platform
IDP

!!! Product

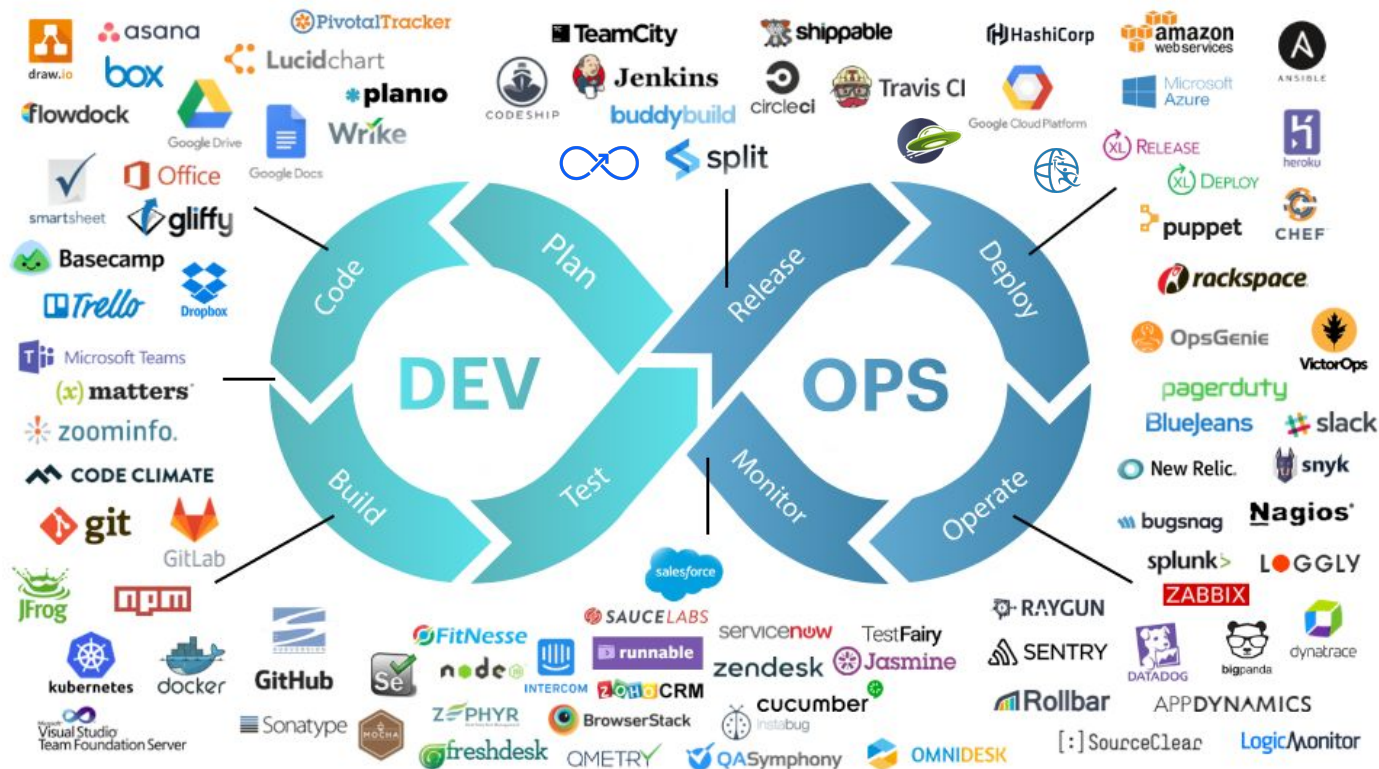
The platform engineering recipe

Ingredients

- Integrated Development Environments
- Version Control Systems
- Resources - containers, databases, storage, compute, transit, etc.
- Runtimes and frameworks
- Artefact, container registry
- Kubernetes
- Observability
- Vulnerability scanners
- CI tools
- GitOps/CD tools
- Infrastructure as Code tools

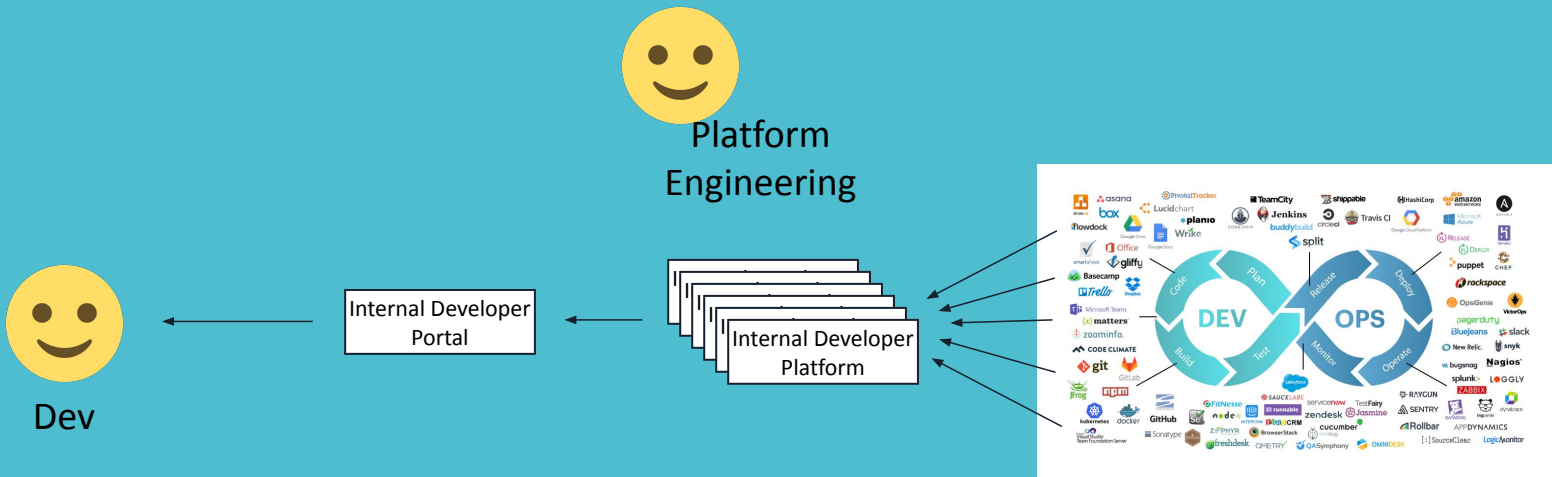


Bake your own platform



Source: <https://www.openxcell.com/devops/>

So, what does it look like?



DEMO: let's configure an AI platform (manually)

URL:

<https://developers.redhat.com>

<https://sandbox.redhat.com>

GitOps

Name	Last commit	Last update
..		
 game-secure-argocd-app-dev-build.yaml	initial commit	41 minutes ago
 game-secure-argocd-app-dev.yaml	initial commit	41 minutes ago
 game-secure-argocd-app-preprod.yaml	initial commit	41 minutes ago
 game-secure-argocd-app-prod.yaml	initial commit	41 minutes ago
 game-secure-argocd-repo.yaml	initial commit	41 minutes ago

game-secure-argocd-app-dev.yaml 1.22 KIB

Blame Edit Replace

```
1 ---
2 apiVersion: argoproj.io/v1alpha1
3 kind: Application
4 metadata:
5   name: game-secure-dev
6   namespace: janus-argocd
7   finalizers:
8     - resources-finalizer.argocd.argoproj.io
9 spec:
10   project: janus
11   source:
12     repoURL: https://gitlab-gitlab.apps.cluster-kgv66.kgv66.sandbox3262.opentlc.com/development/game-secure-gitops.git
13     targetRevision: HEAD
14     path: ./helm/app
15     helm:
16       parameters:
17         - name: namespace.name
18           value: game-secure-dev
19         - name: environment
20           value: dev
21         - name: image.tag
22           value: latest
23     destination:
24       server: https://kubernetes.default.svc
25       namespace: game-secure-dev
26   ignoreDifferences:
27     - group: '*'
28       jqPathExpressions:
29         - '.imagePullSecrets[] | select(.name | contains("-dockercfg-"))'
30     Kind: ServiceAccount
31   syncPolicy:
32     automated:
```

... Environment definitions
are stored and managed
in version control.
(and then applied)



Recap

- Terminology
 - Platform engineering
 - Cognitive load
 - GitOps
- Security & resilience challenges
- Challenges with uncontrolled (SaaS?) environments
- It is not straightforward to enable AI in production
 - ⇒ Stop "playing" with AI, make it enterprise-grade





Challenges

Challenges

Which AI architecture?



coding in 2022

JS stringUtils.js x

```
1 function startsWithCapitalLetter(str) {  
2   return str.length > 0 && str[0] === str[0].toUpperCase();  
3 }
```

coding in 2027

JS stringUtils.js x

```
1 import Anthropic from '@anthropic-ai/sdk';  
2  
3 const anthropic = new Anthropic({  
4   apiKey: process.env.ANTHROPIC_API_KEY,  
5 });  
6  
7 export async function startsWithCapitalLetter(str) {  
8   const prompt = `You are a capitalization expert.  
9     You have been studying capitalization for years.  
10  
11     Text to analyze: "${str}"  
12  
13     Respond with ONLY 'true' or 'false'.  
14     Is the first letter of the text a capital letter?`;  
15  
16   const response = await anthropic.messages.create({  
17     model: 'claude-mythos-1-20261225',  
18     max_tokens: 5000000,  
19     temperature: 0,  
20     thinking: 'ultramaxmegathink',  
21     messages: [{ role: 'user', content: prompt }],  
22   });  
23  
24   const answer = response.content[0].text.trim().toLowerCase();  
25   return answer === 'true';  
}
```

achines
at require

eing

il networks.

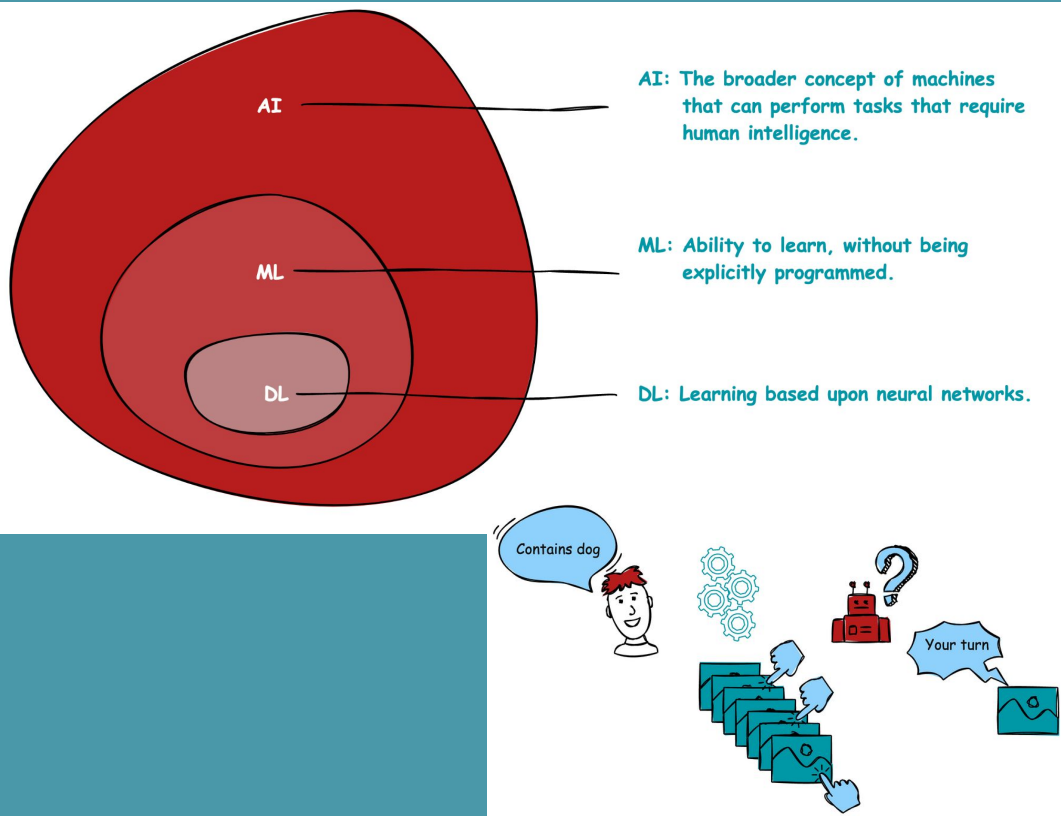


... Don't we
default to LLMs?



Blog
http
http://youtube.com/10410/road-ai-guide-understanding-ai-ml-models

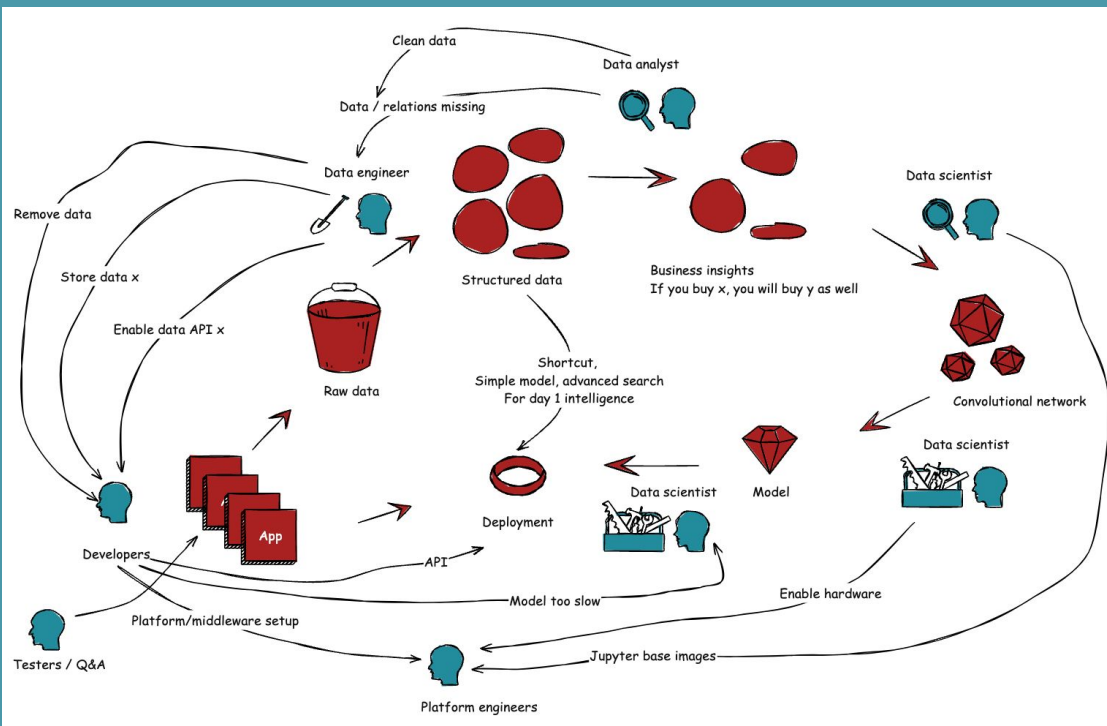
104/10/road-ai-guide-understanding-ai-ml-models



Blog:

<https://developers.redhat.com/articles/2025/04/10/road-ai-guide-understanding-ai-ml-models>

<https://youtu.be/-TG3gbSeVGO>



Challenges

Which LLM-based
architecture?



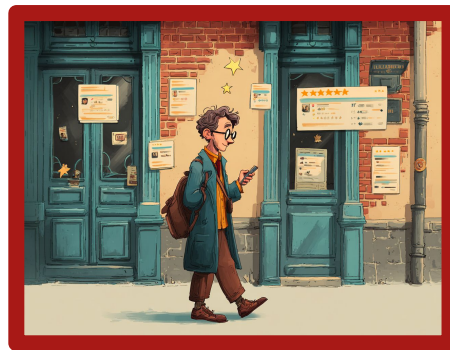
Three Friends going to a bar..



Fine-Tuna



Agent Gaga

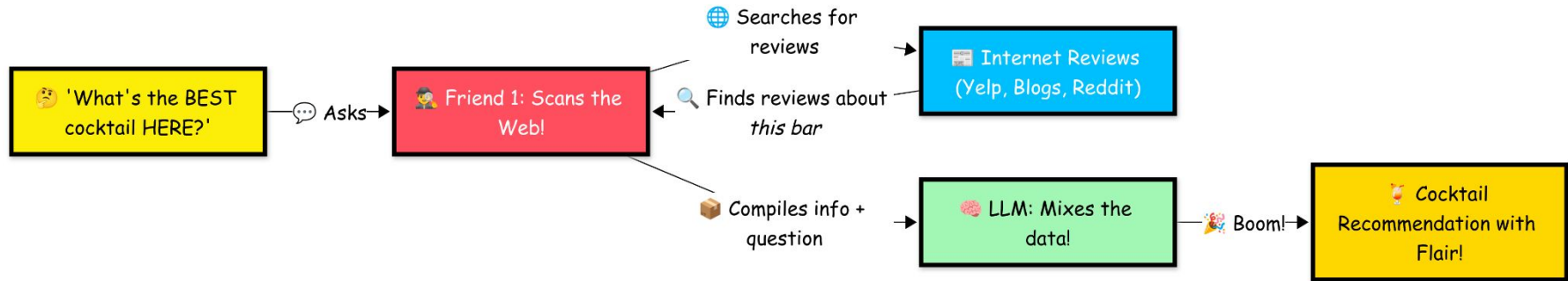


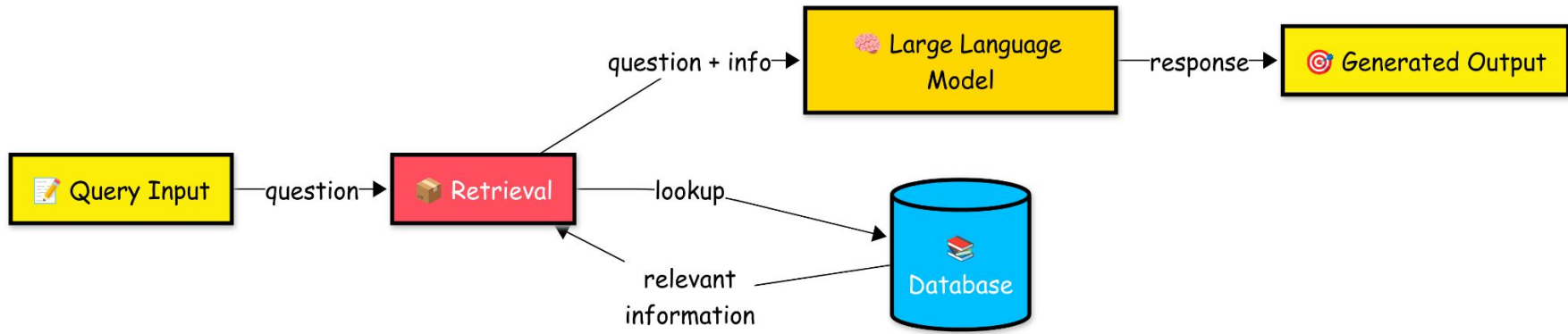
Retrievee

Meet Retrievee

- Immediately pulls out phone.
 - Orders based on a 4.3-star review from 2017.
- Knows nothing, but finds everything .







Meet Fine-Tuna

- Always orders the same thing.
 - Panics if the menu changed.
- Memory-based. No improvising.



RAG vs Fine-Tuning



	RAG	Fine-tuning
External Access	Yes	No
Response Time	Slow	Fast
Knowledge Update	Easy	Difficult
Adaptability	High	Lower
Inference Cost	Higher	Lower



InstructLab

Meet Agent Gaga

- Already talked to the bartender.
- Pre-ordered your drinks.

→ Combines memory, research, and vibes.

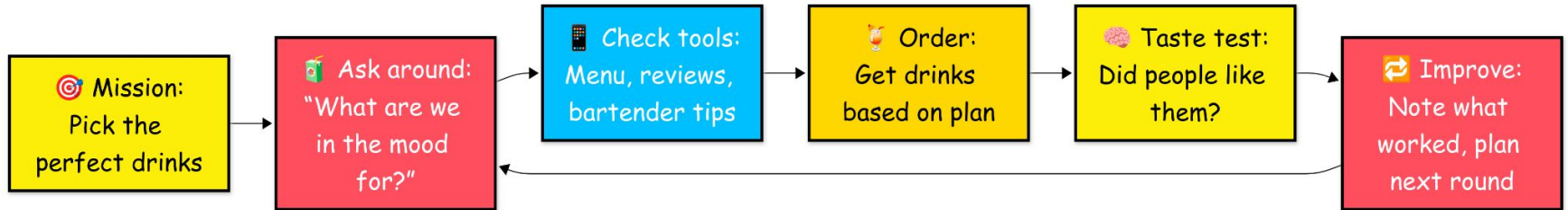


MCP



**One plug
for every tool.**





The risks

Lack of transparency
→ No clear accountability



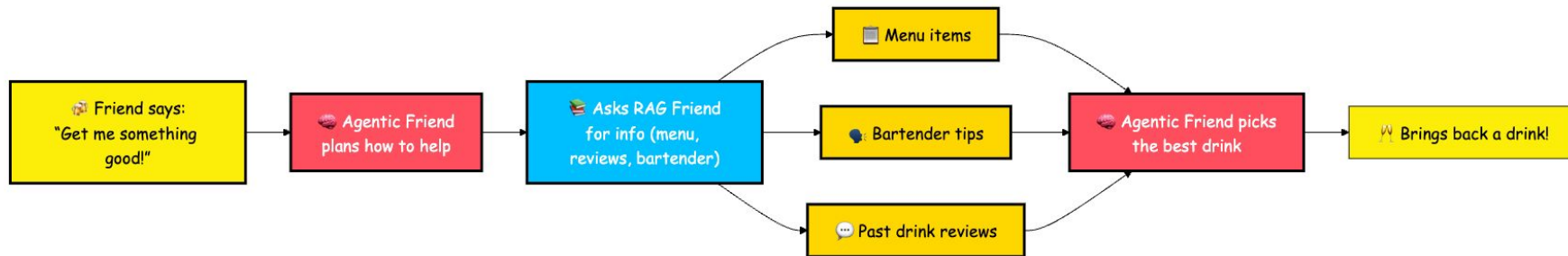
Agentic + RAG



+



AGENTIC RAG



DEMO: InstructLab

URL: <https://interact.redhat.com/share/YTOLvtIRWFuucuvxKf7D>

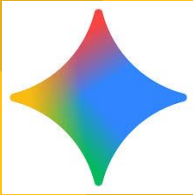
DEMO: Quarkus AI

URL: https://redhat-developer-demos.github.io/quarkus-tutorial/quarkus-tutorial/17_ai_intro.html

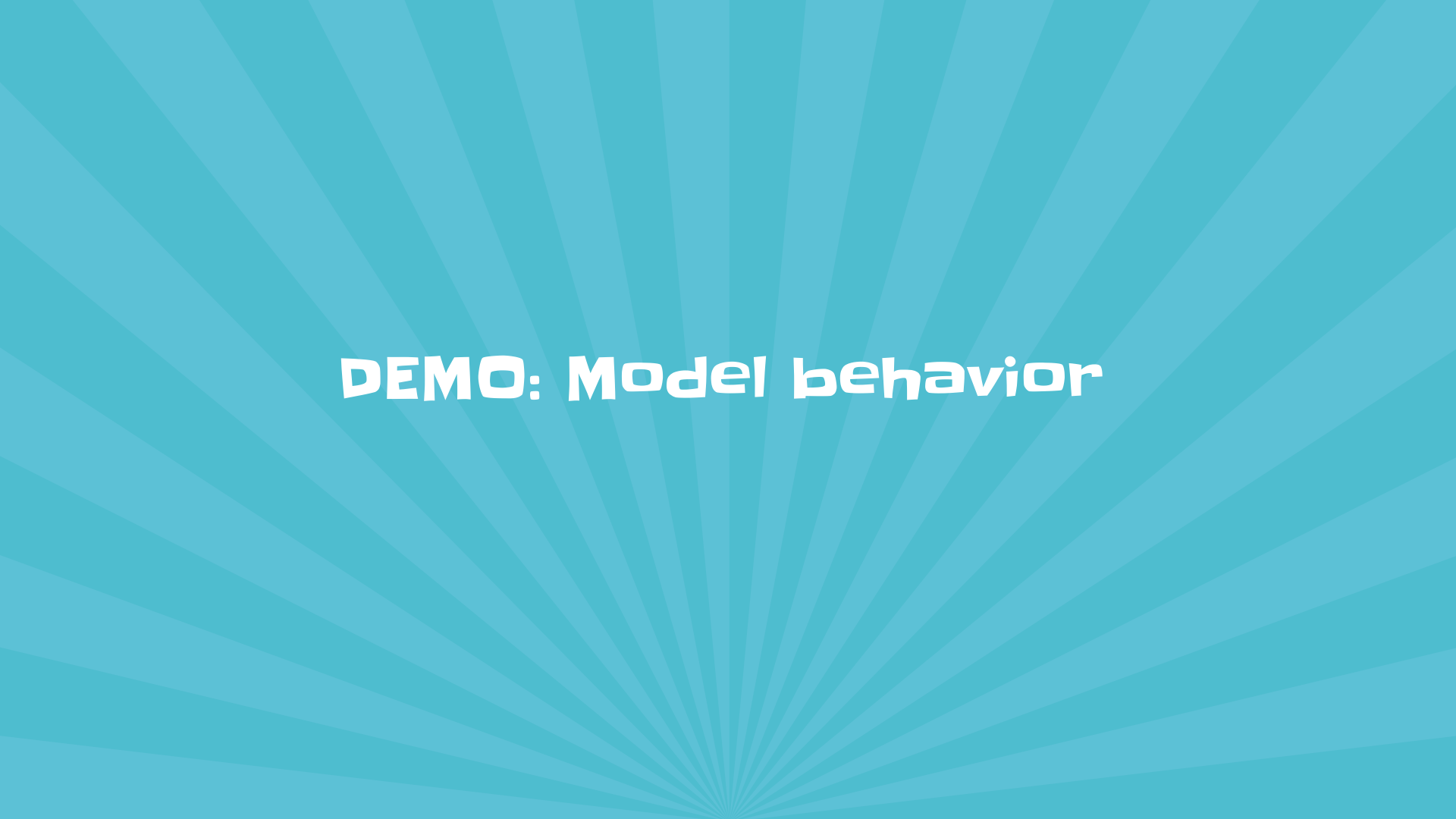
Challenges

Which models?





... Where to start?
What is that (IBM)
Granite model
About? Does size
matter?



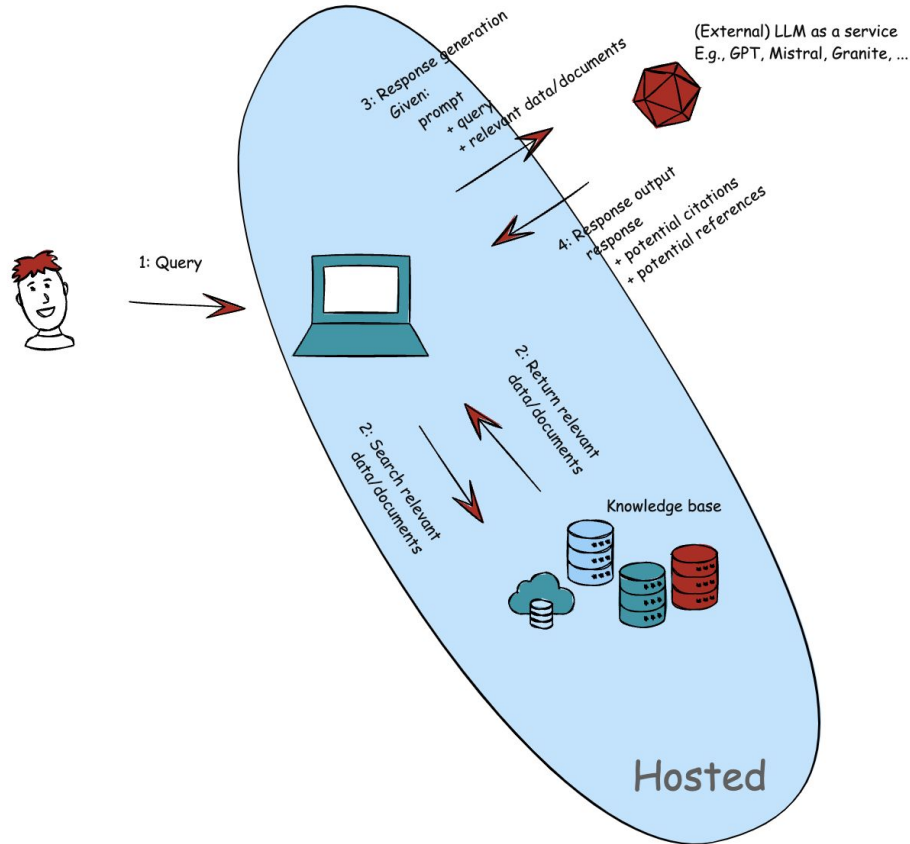
DEMO: Model behavior

Challenges

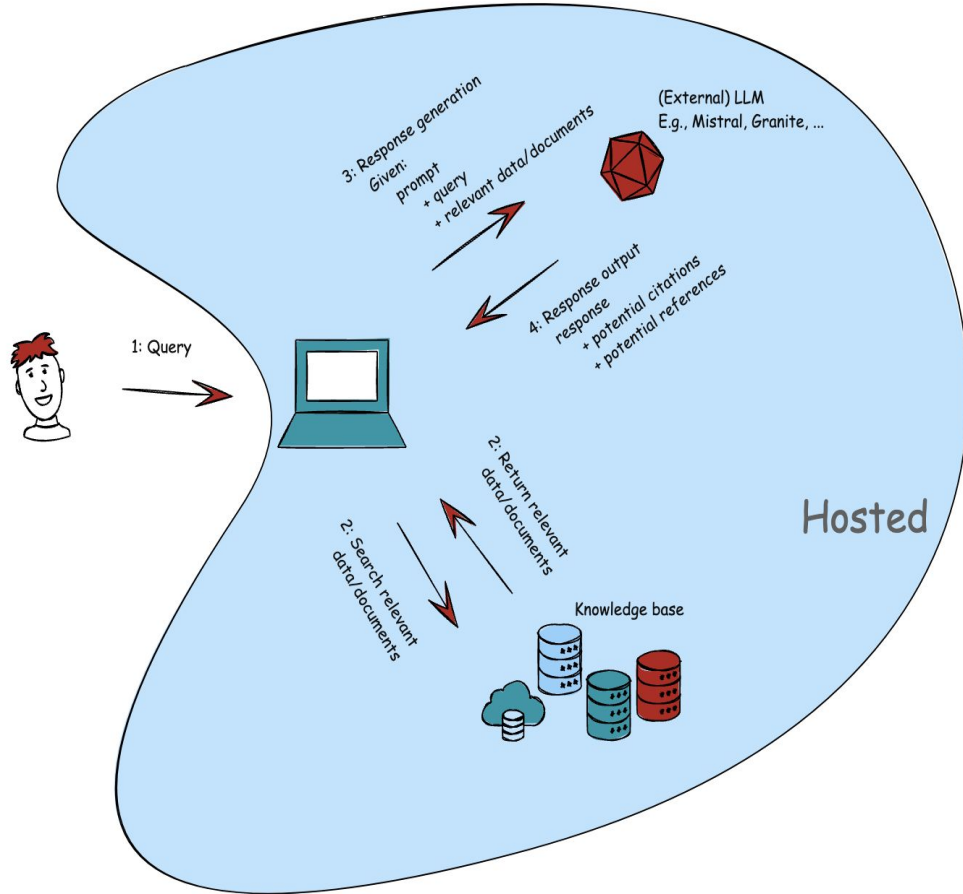
What about data flows?



RAG with uncontrolled model



RAG with controlled model



Challenges

Sovereignty



What the CLOUD Act Really Means for EU Data Sovereignty

The CLOUD Act allows U.S. authorities to access data stored in the EU, putting it in direct conflict with GDPR. Learn how this impacts data sovereignty and what EU businesses can do to stay compliant



Wire
Jul 15, 2025



[REDACTED]: EU data not protected from US access

21 July, 2025 11:11:35



This article is also available in German.

Explosive testimony before the French Senate: Microsoft cannot guarantee that European user data is safe from US authorities despite EU hosting.

[REDACTED] cannot protect European user [REDACTED] access by US authorities – even if it is stored in EU data centers. This was admitted under oath by the legal director of [REDACTED] France, [REDACTED], at a [hearing before the French Senate](#) on June 10, 2025.

... Is sovereignty only about not trusting the barkeeper?



#	Sovereignty Objectives	Weight in Scoring
SOV-1	Strategic Sovereignty	15%
SOV-2	Legal & Jurisdictional Sovereignty	10%
SOV-3	Data & AI Sovereignty	10%
SOV-4	Operational Sovereignty	15%
SOV-5	Supply Chain Sovereignty	20%
SOV-6	Technology Sovereignty	15%
SOV-7	Security & Compliance Sovereignty	10%
SOV-8	Environmental Sustainability	5%
Total		100%

Twee problemen die ik voorzie

- Relatief korte levensduur van bepaalde modellen gaat ervoor zorgen dat we heel veel test werk gaan moeten doen wanneer we ergens in onze processen een LLM gebruiken dat op zogenaamd “pensioen” gaat.
 - Kan Redhat ervoor zorgen dat deze modellen ergens autonoom kunnen blijven doordraaien? Als ze op een bepaald moment in de tijd goed genoeg waren voor onze use-case, dienen we deze niet noodzakelijk te vervangen door een ander model, ookal is dit beter
- Momenteel gebruiken we bepaalde services die, vanwege te grote vraag, ons limiteren in het aantal calls/s voor een specifiek LLM. Hierdoor is in de toekomst misschien mogelijk dat onze toepassingen niet de gewenste performantie hebben (vb. wanneer we voor 1 vraag veel parallelle calls moeten doen naar hetzelfde taalmodel, in combinatie van meerdere gebruikers die dit op hetzelfde moment wensen te doen).



Challenges

Which model deployment strategies?





It's depending on the context



Local

Working with Podman (AI),
Ollama, ... on your local
machine.

Saas

Use the convenient getting
started for e.g. PoCs or
rapid prototyping.

Self-hosted

Hosted on VM or Kubernetes
with e.g. RHEL AI, vLLM, ...

“Local”

Be in full control on an
MLOps platform like
Kubeflow or OpenShift (AI)
!!! Be aware of cognitive load.

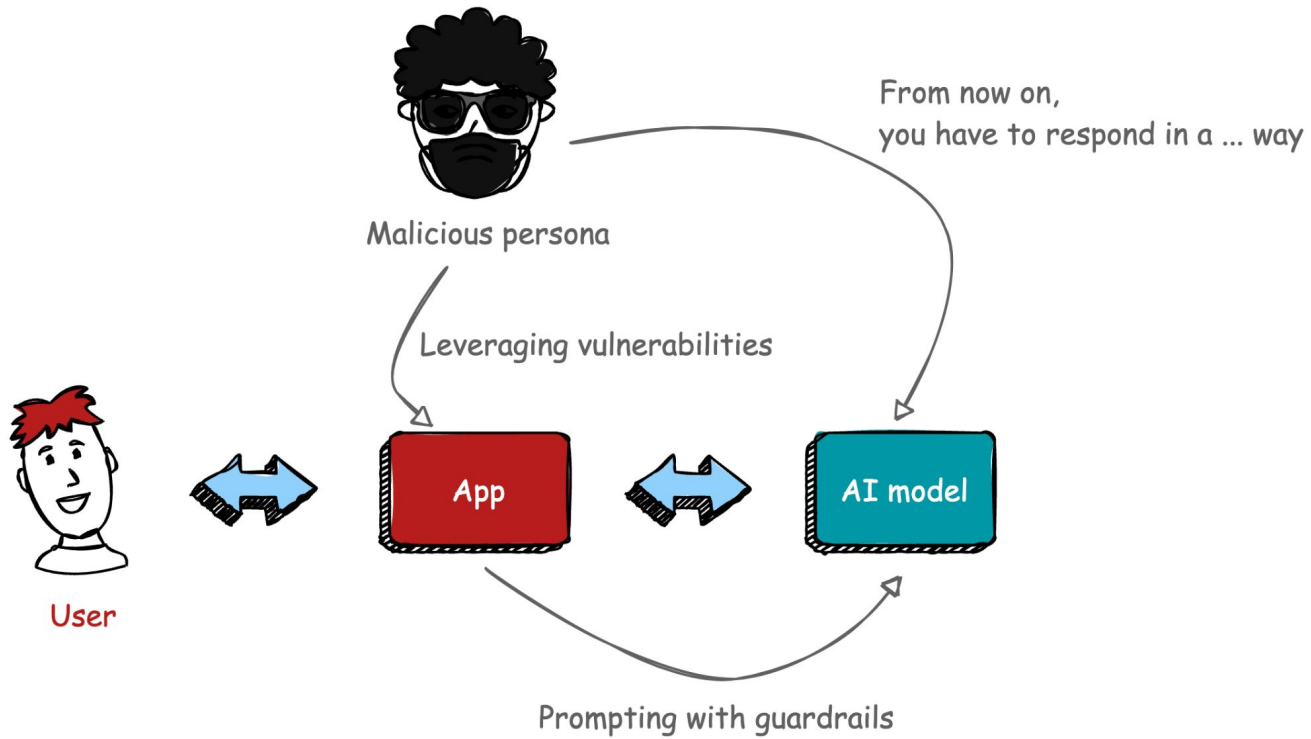
**“Keep your
options open!”**



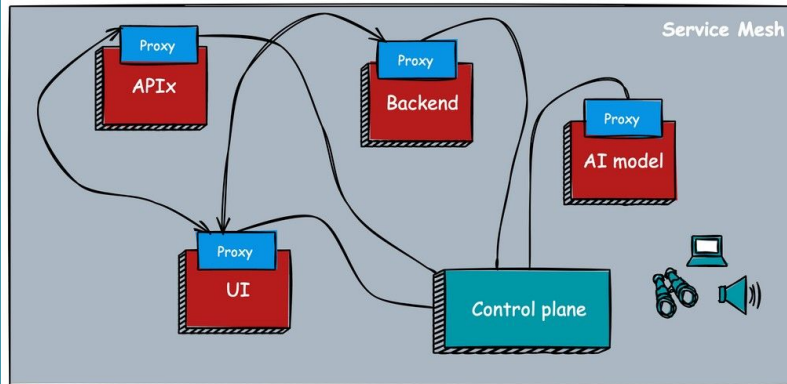
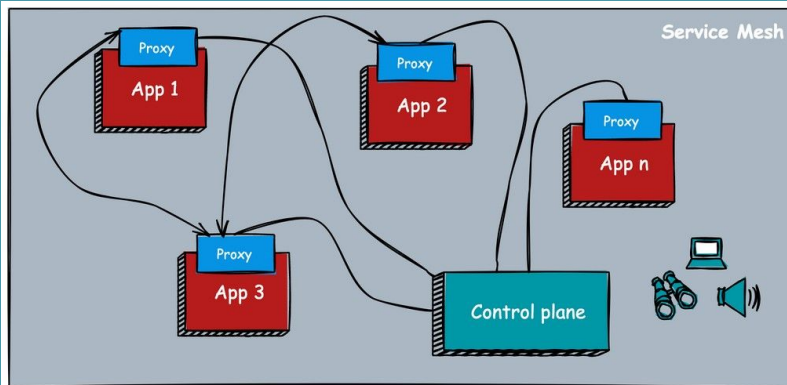
Challenges

What about model
security?





Service mesh



Communication: App x can communicate with App y within the service mesh

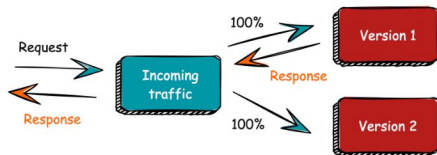
... Centralized management through proxies



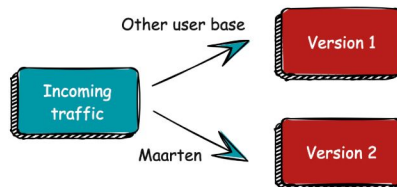
Waiters, bartenders, manager to pay the bill $\Rightarrow$ all working with mobile apps

Service mesh

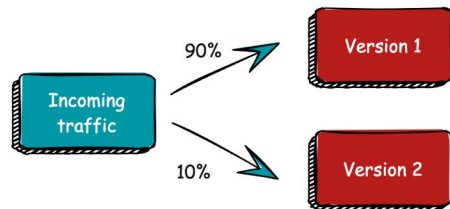
Mitigate deployment risks in a distributed application environment



Validation - Mirroring



Testing - Canary releases



Rolling out - Blue/green deployments

... Can't we use these deployment patterns to validate e.g. models?



Waiters, bartenders, manager to pay the bill $\Rightarrow$ all working with mobile apps

The background of the slide features a series of light blue radial lines that originate from a single point at the bottom center and fan outwards towards the top edges, creating a sunburst or starburst effect.

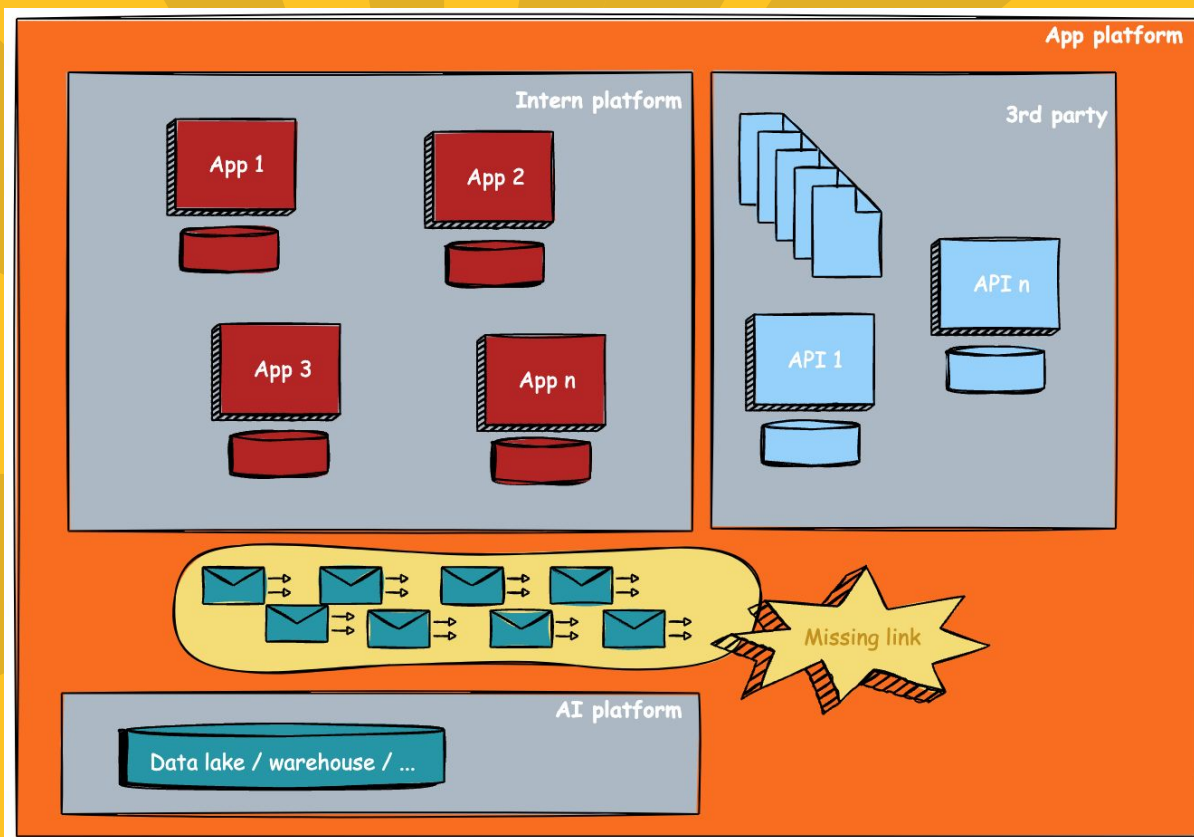
DEMO: service mesh

Challenges

What about data
platforms?



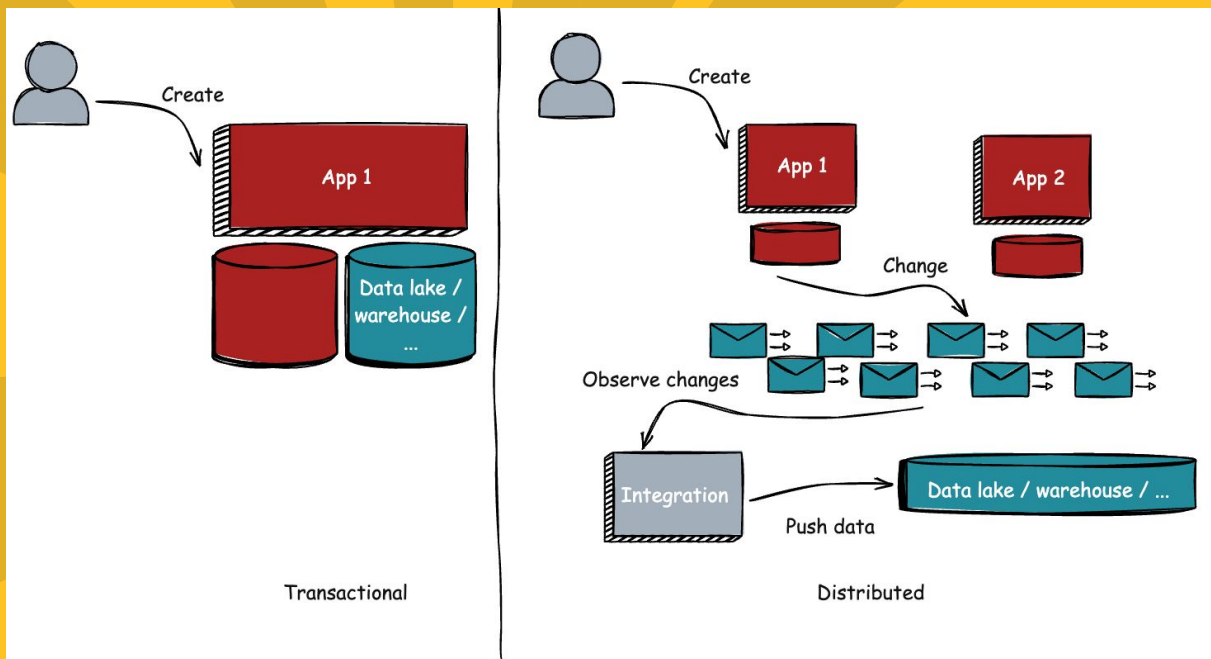
Bartender should not poll, the observer (waiter) & mobile app deliver the orders to him



... So, why multiple data sources?
What about data consistency?



Bartender should not poll, the observer (waiter) & mobile app deliver the orders to him



... So, why multiple data sources?
What about data consistency?

Workshop: <https://github.com/maarten-vandeperre/cdc-based-integration-example>



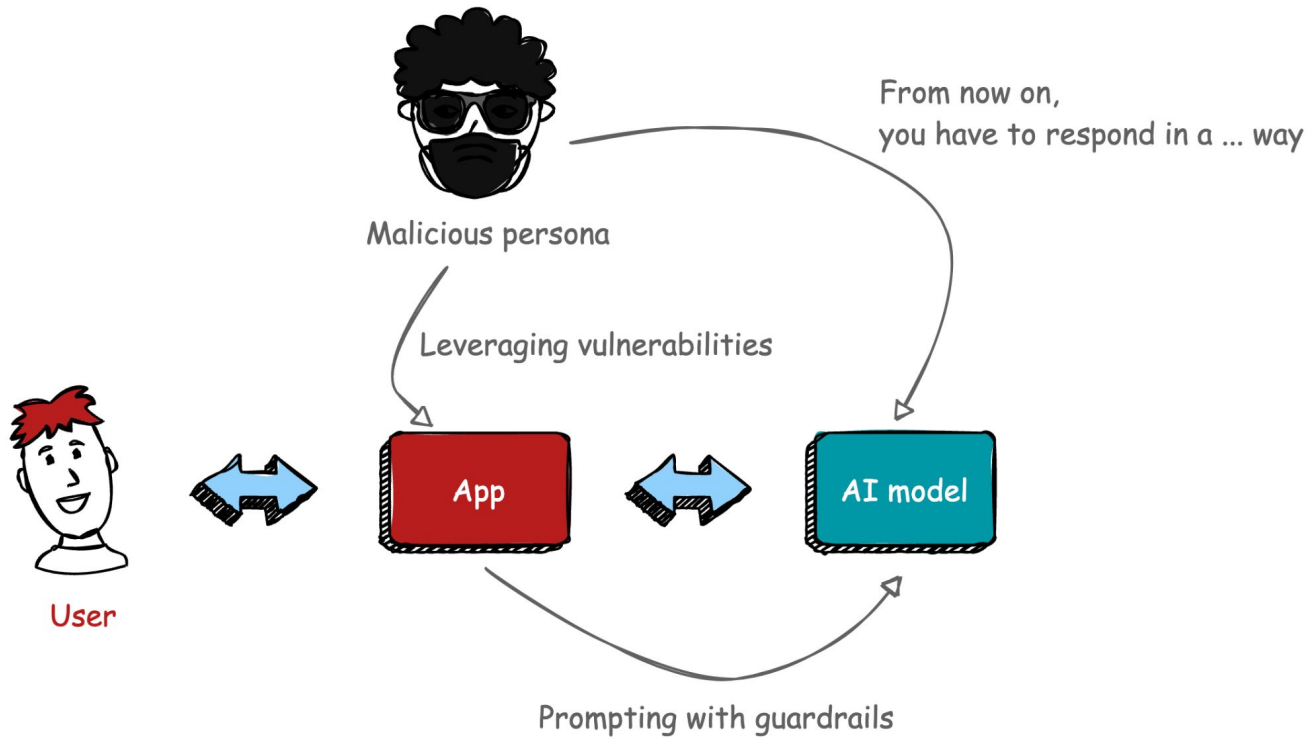
DEMO: CDC

URL: <https://github.com/maarten-vandeperre/cdc-based-integration-example/tree/main>

Challenges

Why do we need
monitoring and
observability?





Chatbot

Hello, how can I help you?

who is kafka?

Franz Kafka was a Czech-language writer known for his surreal and existential works that explore themes of alienation and bureaucracy.

and how is he related to AI? e.g., emergency stop

I do not know who this is. Try again!

who is kafka? and how is he related to ai?

Kafka is a software framework that acts as the backbone of agentic AI, facilitating real-time data processing and communication.

who is kafka? And how does Kafka and a service mesh help with AI?

Kafka is a distributed streaming platform that can help agentic AI by managing real-time data flow, enabling efficient communication and processing between microservices in a service mesh architecture.

Demo video:

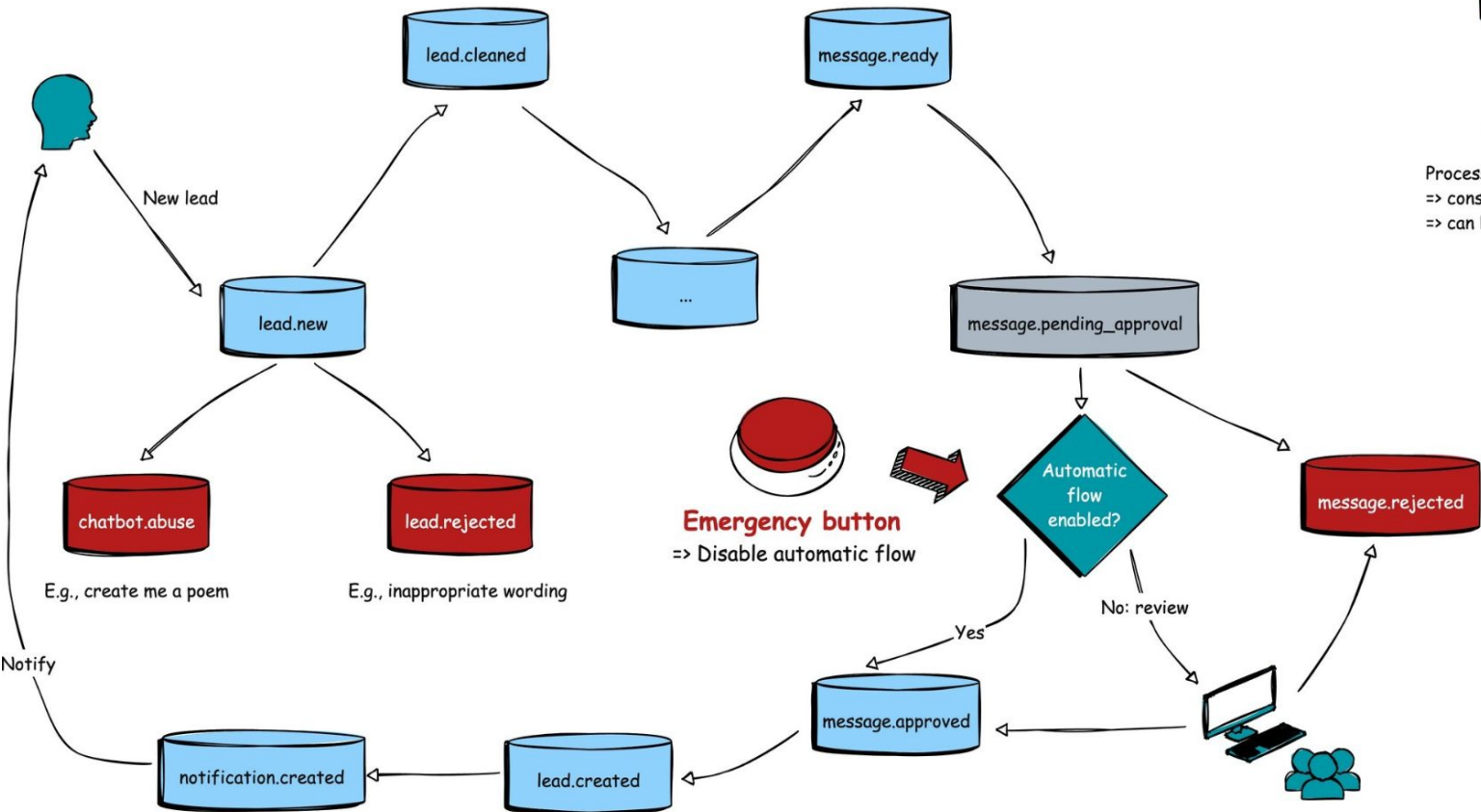
<https://youtube.com/watch?v=j5HuRWWi5mA&si=uFMAAn7EPIMPLqJHg>



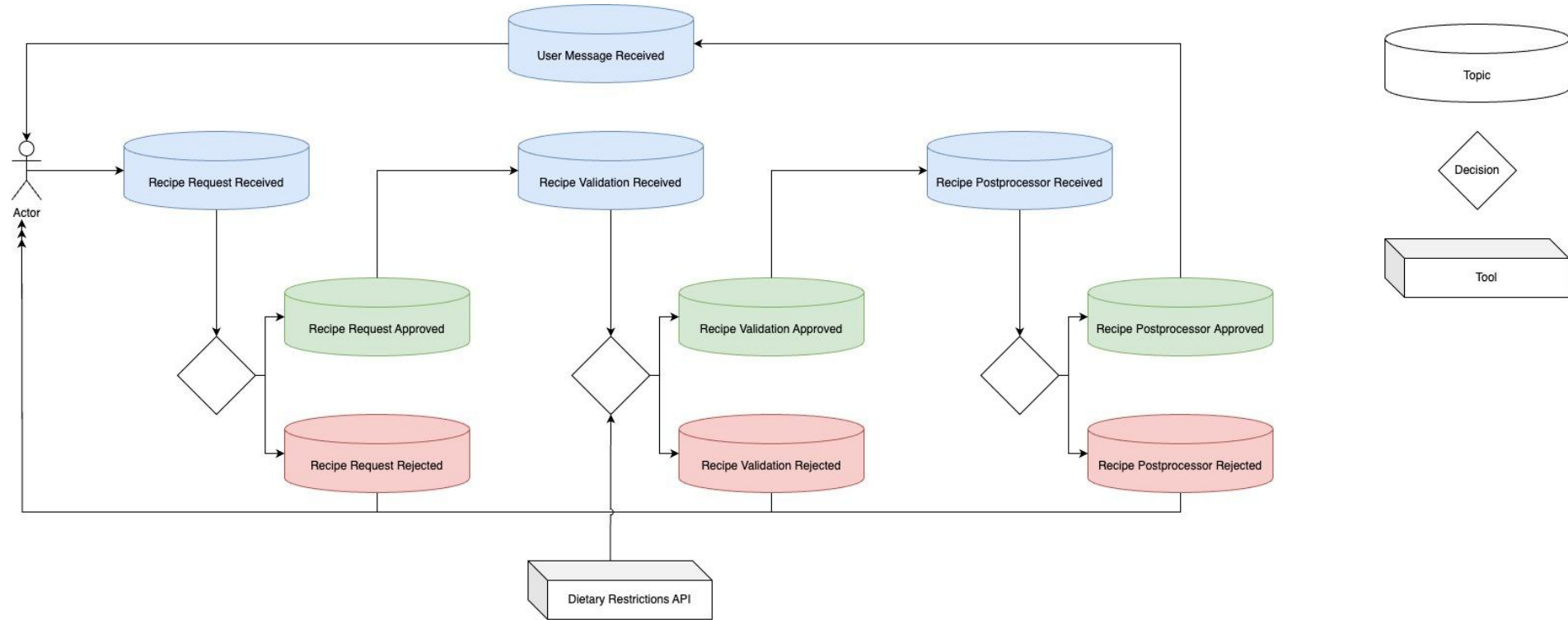
Kafka topic



Processors (e.g., Camel/Knative/...)
=> consumer + processing + producer
=> can be API, MCP, LLM, database integrated



Emergency button
=> Disable automatic flow



Demo video:

<https://youtube/j5HuRWWi5mA?si=uFMA7EPIMPLqJHg>

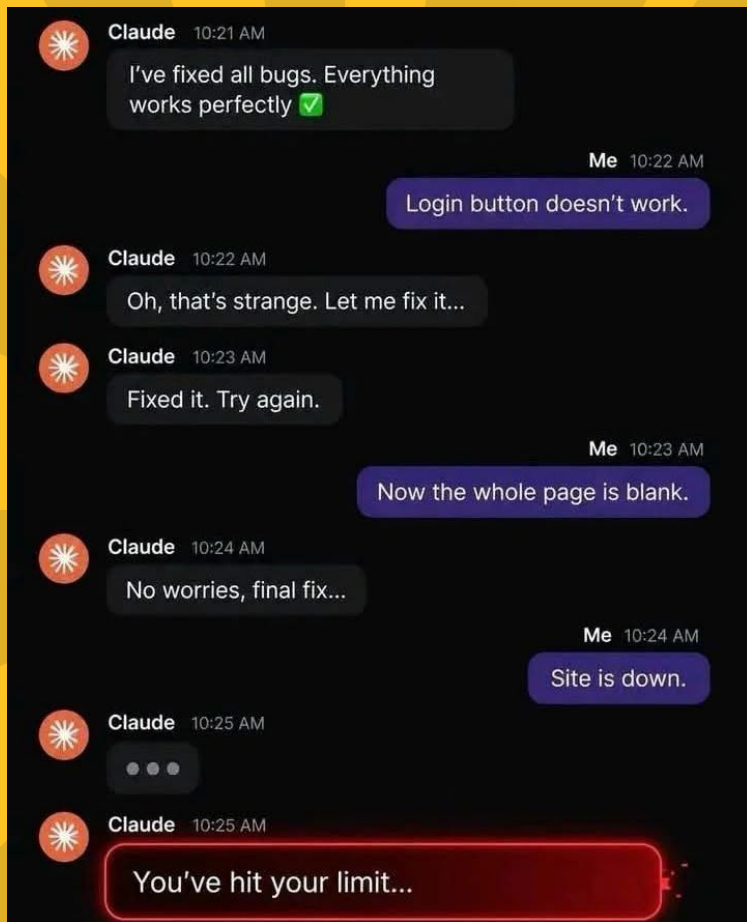
DEMO: Guardrailing

URL: <https://github.com/rh-ai-quickstart/lemonade-stand-assistant/blob/main/docs/images/grafana-dashboard.png>

Challenges

What about vibe coding?





Source: <https://www.linkedin.com/feed/update/urn:li:activity:7457109036631482368/>

Multi-agent:

<https://github.com/luma-repositories/leo-by-lutske-and-maarten/tree/07-multiagent-refactor-to-clean-architecture-backup-qwen/opencode/mode>



```

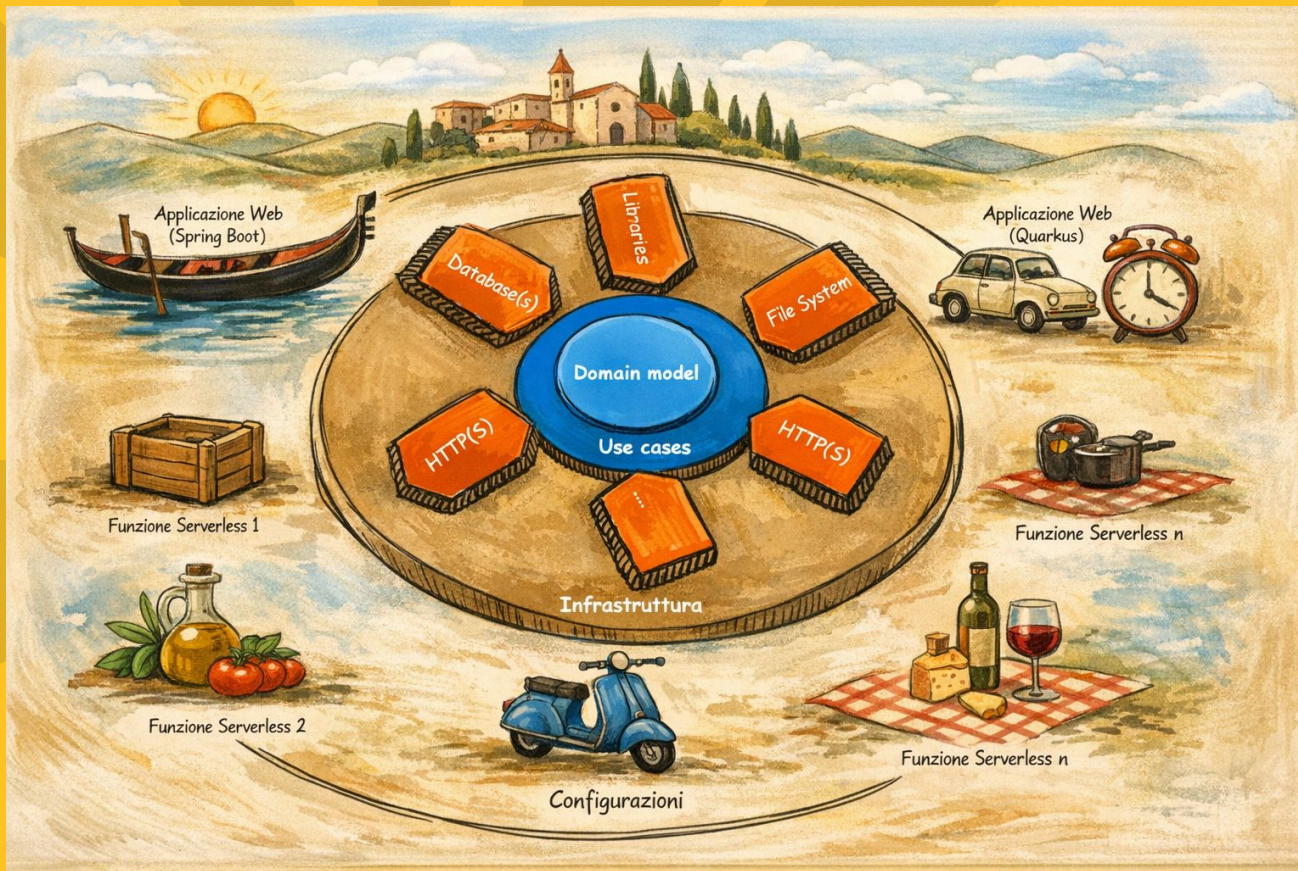
✓ logging
  > _documentation
  ✓ _prompts
    ✓ 01_raw
      .gitkeep
      M+ 20260416_1559_01_clean_up_initial_setup.md
      M+ 20260418_1333_01_filter_and_sorting.md
      M+ 20260420_0844_02_only_create_and_delete_for_logging_microservice.md
    > 02_curated
    > 03_approved
    > 04_wip
    ✓ 05_executed
      .gitkeep
      M+ 20260416_1554_01_initial_setup.md
      M+ 20260416_1559_01_changelog.md
      M+ 20260416_1559_01_clean_up_initial_setup.md
      M+ 20260418_1333_01_filter_and_sorting.md
      M+ 20260420_0844_02_only_create_and_delete_for_logging_microservice.md

```

Should I only stick with
the fun, or should I look
into the boring part too?

⇒ Iterate





Well... that's a pity...
Should I only rely
on foundation
models?





Source: <https://drinkotron.com/memes/drinkotron-drunk-forty-40-years-hat-grandma-meme>

... Why the ~~xxx~~?
How is this
related to vibe
coding?



TODO - Demo checklist

What should be demoed:

- Trusted software supply chain

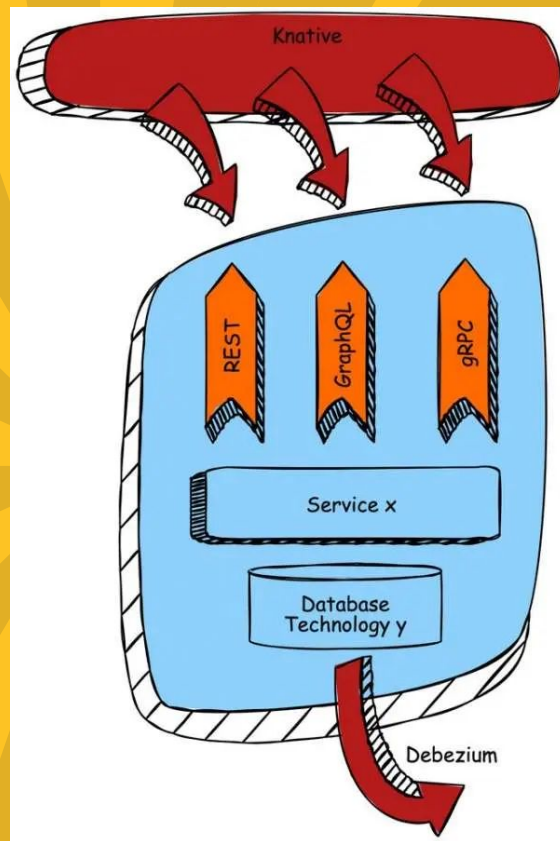


Challenges

What about compliance
and standardization?



I don't want to visit the same pub, drink the same drinks every day.



... I don't need
standardization: it
blocks innovation!



TODO - Demo checklist

What should be demoed:

- Trusted software supply chain
- Templates - GitOps



Challenges

What about the cognitive
load?



TODO - Demo checklist

What should be demoed:

- Trusted software supply chain
- Templates - GitOps
- Centralized (software lifecycle)
- Abstractions to reduce cognitive load



DEMO: Platform engineering

URL:

<https://developers.redhat.com/learn/deploying-and-troubleshooting-red-hat-developer-hub-openshift-practical-guide>

<https://developers.redhat.com/learn/streamline-development-github-integration-and-software-templates-red-hat-developer-hub>

At:

(!!! Use the openshift-ai/4_Models_inferencing folder)

<https://developers.redhat.com/learn/openshift-ai/build-and-evaluate-fraud-detection-model-tensorflow-and-onnx>

Summary

You can find my slides, recordings, and blog posts at maarten-vandeperre.be.
And feel free to share feedback & possible improvements!

Context	Starting with AI is easy, getting it in production can be challenging .
Platform engineering	Can reduce the cognitive load via abstraction layers, self-service and automation on a solid platform (for #personas within your IT-landscape). E.g., OpenShift/Developer Hub.
GitOps	Can bring and standardization while maintaining flexibility for dev teams.
Challenge: which (AI) architecture	Start with use cases, SLAs, ... Don't default to AI, because you want to work with AI, go for AI when it's the best suited implementation detail . ⇒ Feel free to reach out for this.
Challenge: what about the data?	Think about change-data-capture , split application data and data science data, think about data confidentiality (i.e., sovereignty), think about hybrid- or multi-cloud . ⇒ Feel free to reach out for this.
Challenge: security, observability, monitoring	Think about defaulting to service meshes within your platforms: it brings centralized mTLS, monitoring, observability, advanced routing, ... ⇒ Feel free to reach out for this.
How to start?	Well, feel free to reach out, go to developers.redhat.com , contact Red Hat within your region for a workshop/introduction, or, again, feel free to reach out.

BARTENDER:

"YOU'VE HAD ENOUGH. NOW GO FOR COFFEE."



**WHEN THE BAR ITSELF INTERVENES,
IT'S TIME TO SWITCH DRINKS.**

Let's switch from
exposing AI to consuming
AI.



The Playbook for a Sovereign MaaS Platform



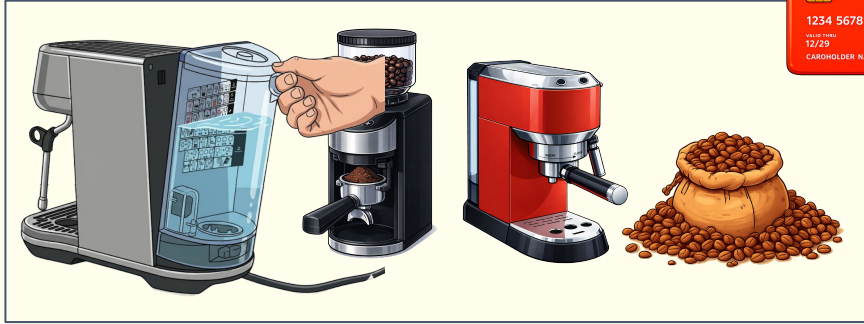
... imagine you need to provide coffee to your employees.



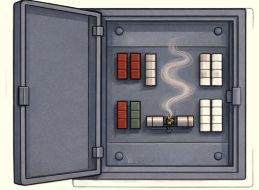
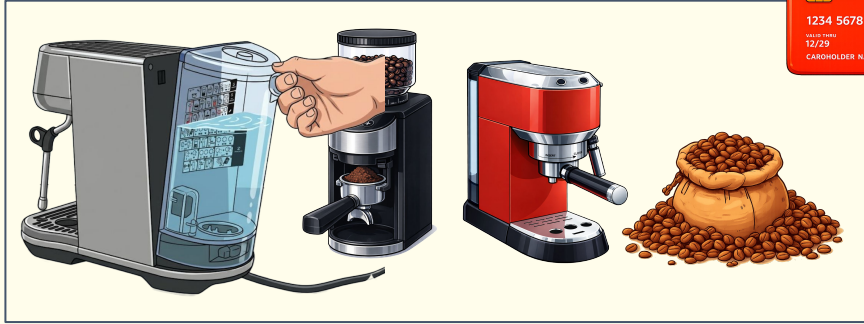
Do It Yourself



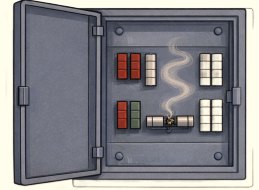
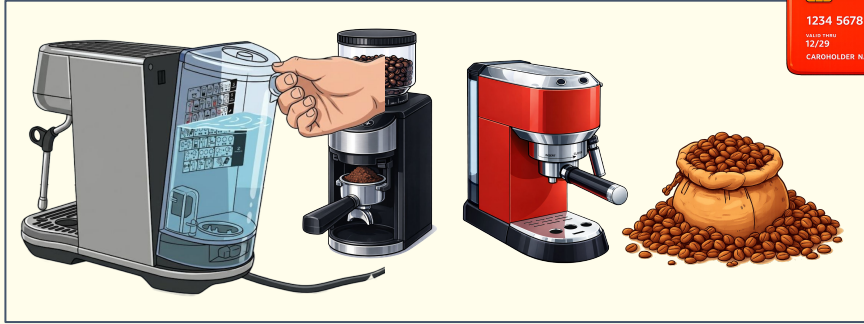
Shadow AI



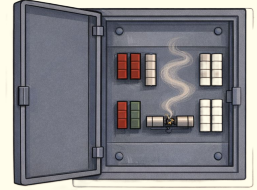
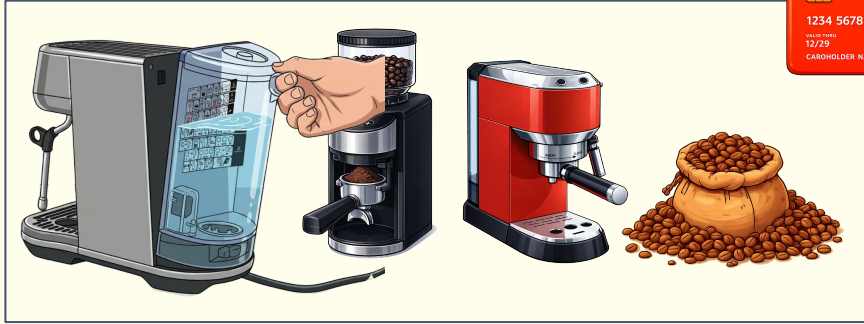
Shadow AI



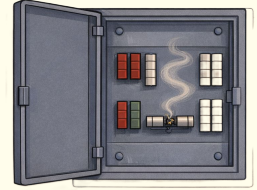
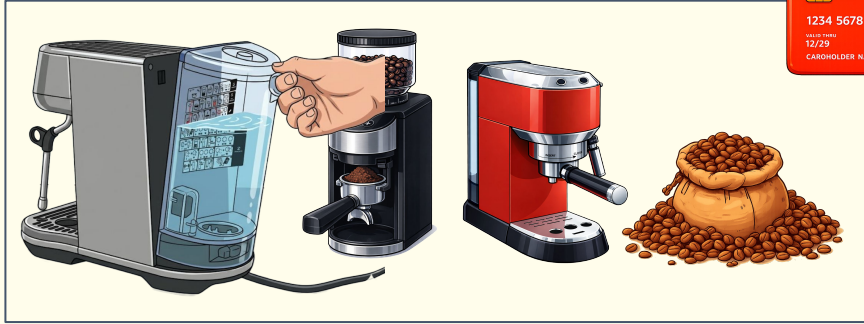
Shadow AI



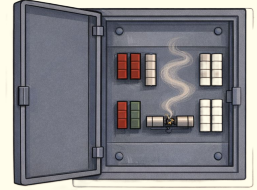
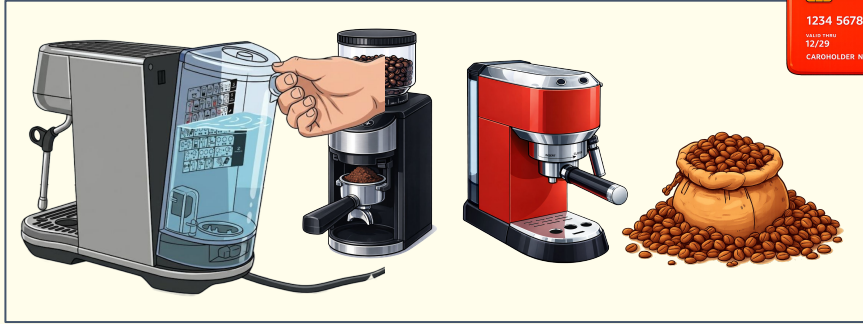
Shadow AI



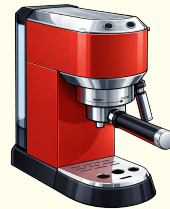
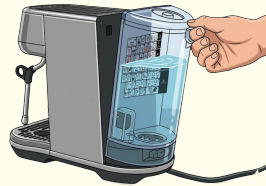
Shadow AI



Shadow AI

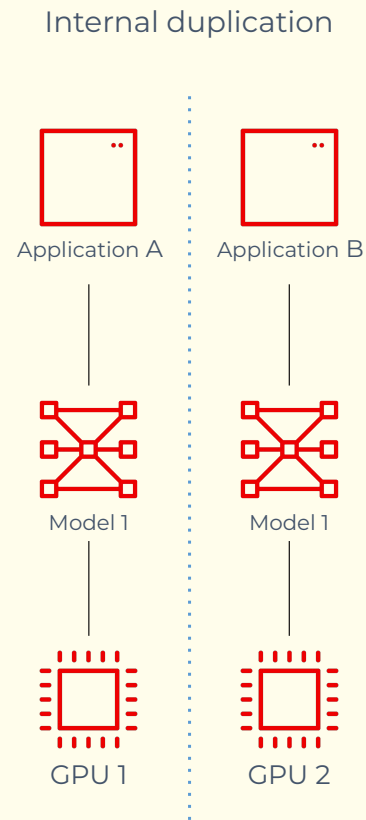


....but why are we talking about coffee?

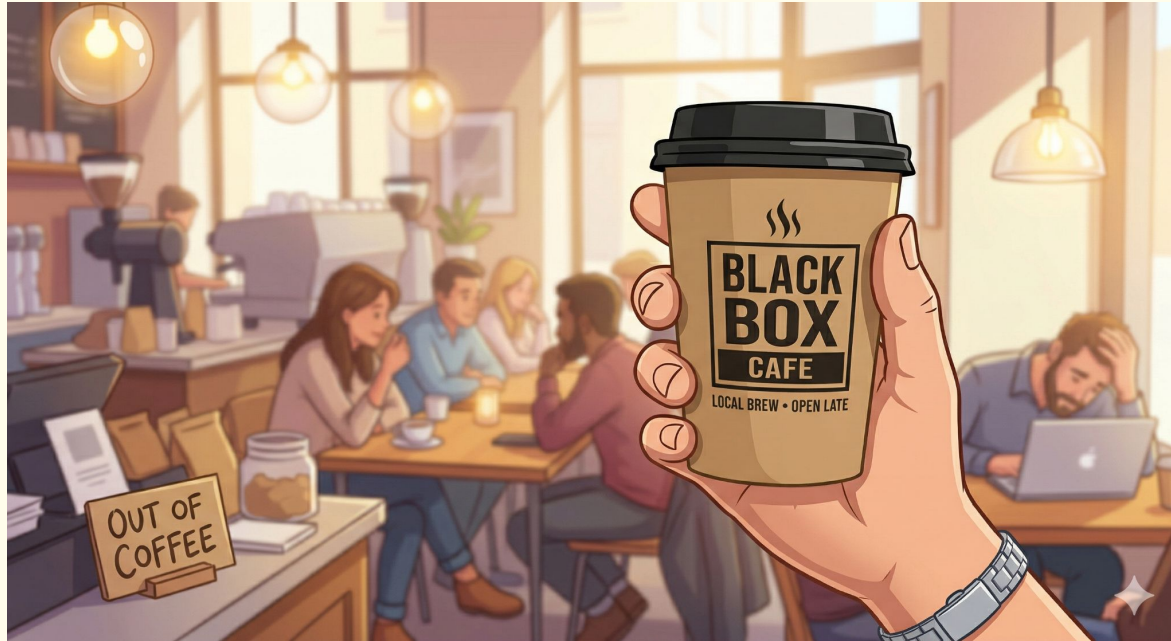


What shadow AI can lead to ...

- Duplicate model deployments
- Fragmented governance
- Resource underutilization



Expense the coffee

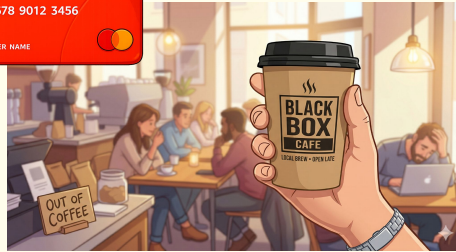


- **The Competitor / External Proprietary API (Slide 6):** Instead of "Expensing the coffee" (which might imply a specific brand), frame it as **"Sending employees to an expensive, external commercial coffee shop."** * *Why this is a great competitor analogy:* It's very easy and the coffee is good, BUT it's expensive, you have no control over price hikes, and most importantly, **data privacy**. The barista at the external shop knows exactly what your employees are working on and drinking.

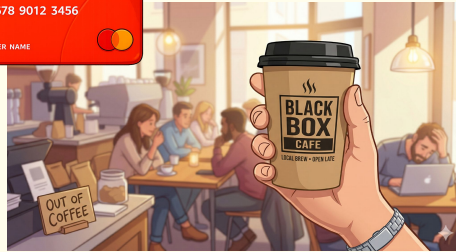
Revamp Slide 7 ("What this can lead to..."): List the exact business pain points of *not* having MaaS:

1. **Runaway Costs:** Unpredictable API bills from external vendors.
2. **Data Leaks / Sovereignty Issues:** Sending company secrets to external coffee shops.
3. **Resource Inefficiency:** Wasting expensive GPUs because teams are hoarding them (everyone having their own desk coffee maker).
4. **Vendor Lock-in:** Being stuck with one menu and one price.

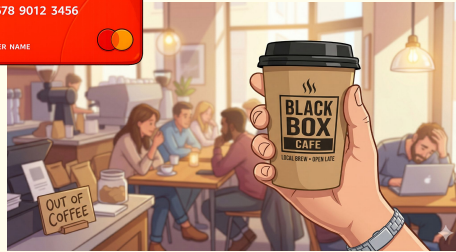
Expense the coffee



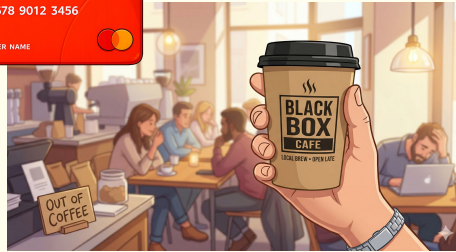
Expense the coffee



Expense the coffee



Expense the coffee



Proprietary Specs?
Giant stacks of
IF-statements.

NO WAY!



What this can lead to ...

- Vendor lock in
- Unpredictable cost
- Less Flexibility (ref. sovereignty)
- Data leakage (ref. sovereignty)



Machine for everyone



Machine for everyone – but limited



Machine for everyone

User using AI app



Prompt

Response

Inference API



Model

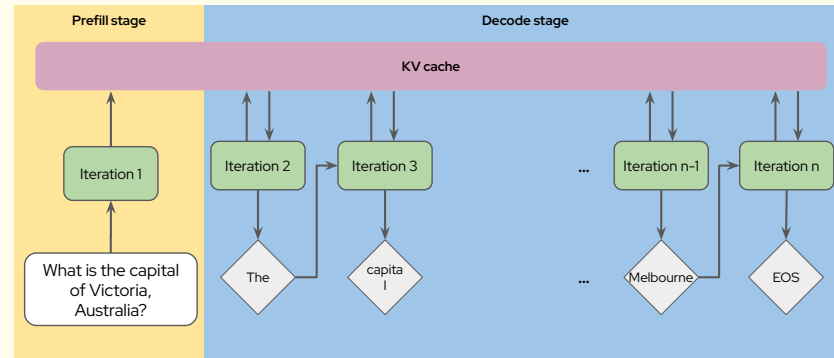


Inference Server



Hardware Accelerator

Process the prompt and generate a response



What are “tokens”?

LLMs consume **tokens**, not characters.

- Segmentation
 - a. Text is chopped into words, syllables, or single characters.
- Mapping
 - a. A model-specific Tokenizer translates these chunks into unique numerical IDs

Qwen/Qwen2.5-72B

Token count
28

"How do I exit Vim? "

"Attached is our 500-page AWS billing report. Please explain why we are broke."

70959, 653, 358, 4869, 94484, 30, 330, 1406, 1, 65987, 374, 1039, 220, 20, 15, 15, 15112, 23245, 33531, 1895, 13, 5209, 10339, 3170, 582, 525, 14422, 48116

Qwen/Qwen2.5-72B

Token count

28

"How do I exit Vim? "

"Attached is our 500-page AWS billing report. Please explain why we are broke."

70959, 653, 358, 4869, 94484, 30, 330, 1406, 1, 65987, 374, 1039, 220, 20, 15, 15, 15112, 23245, 33531, 1895, 13, 5209, 10339, 3170, 582, 525, 14422, 48116

google/gemma-7b

Token count

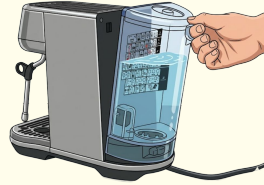
33

"How do I exit Vim? "

"Attached is our 500-page AWS billing report. Please explain why we are broke."

2, 235281, 2299, 749, 590, 13581, 159128, 235336, 664, 108, 108, 108, 235281, 37539, 603, 1167, 235248, 235308, 235276, 235276, 235290, 3323, 42556, 54972, 3484, 235265, 5651, 10200, 3165, 783, 708, 15203, 1816

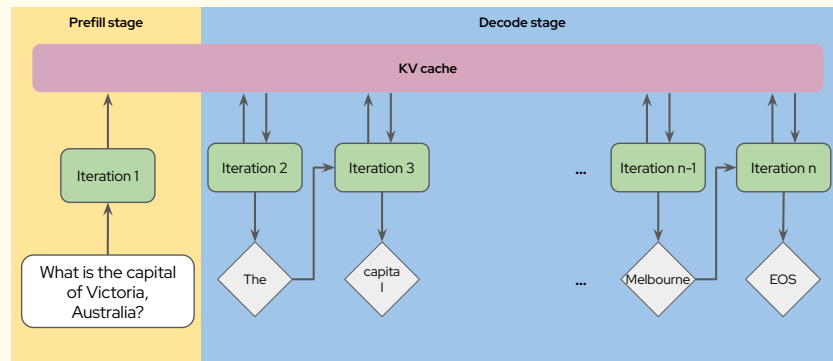
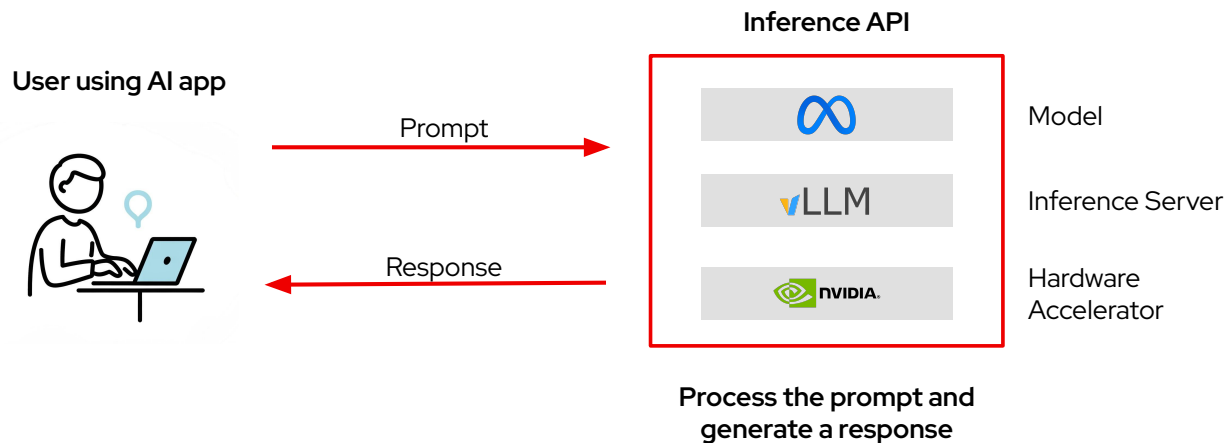
The prompt



"How do I exit Vim? "

"Attached is our 500-page AWS billing report. Please explain why we are broke."

Machine for everyone



The inference engine



The inference engine



Compresses the model so it can run locally

→ built to brew exactly one cup of coffee
for one developer at a time.

The inference engine



Compresses the model so it can run locally

→ built to brew exactly one cup of coffee
for one developer at a time.



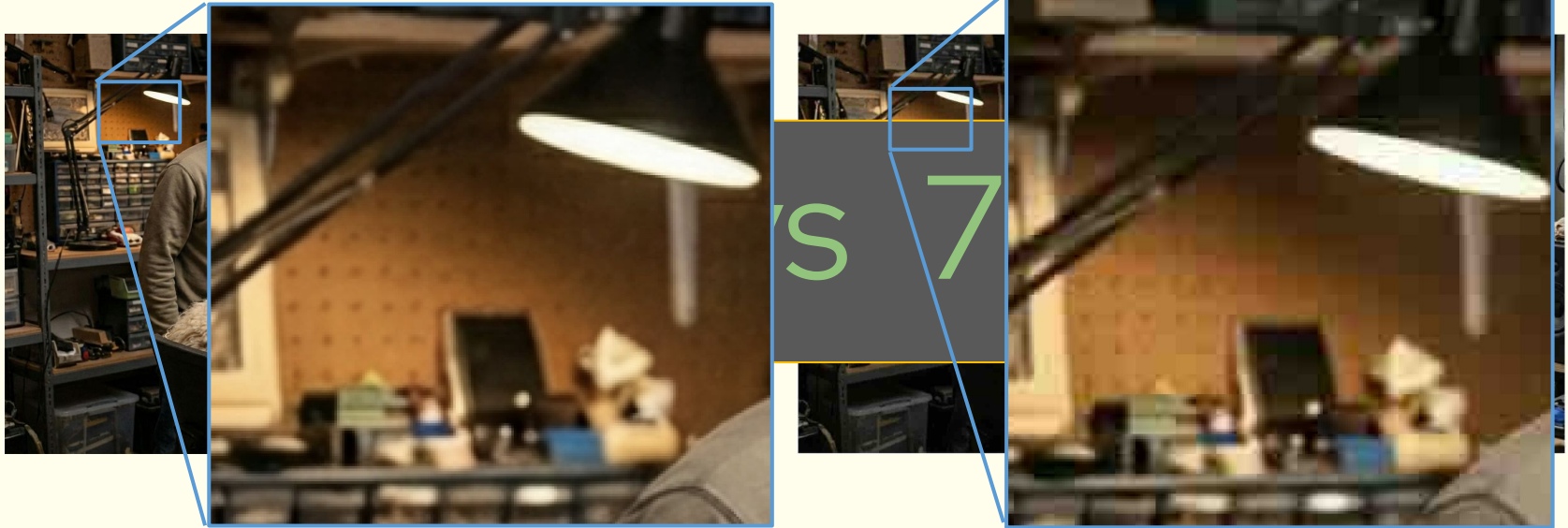
Engine powering modern AI

Able to pull fifty shots of espresso
/requests simultaneously.

...but why quantization?

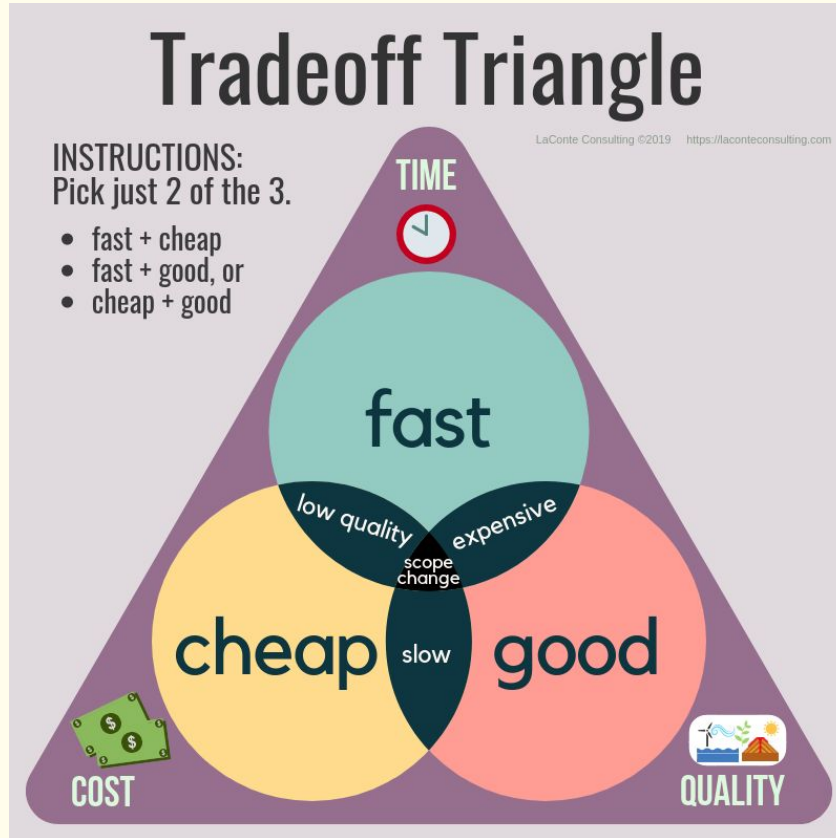
uncompressed - original size

compressed - quantised



Can you see the difference?

Remember the Tradeoff Triangle

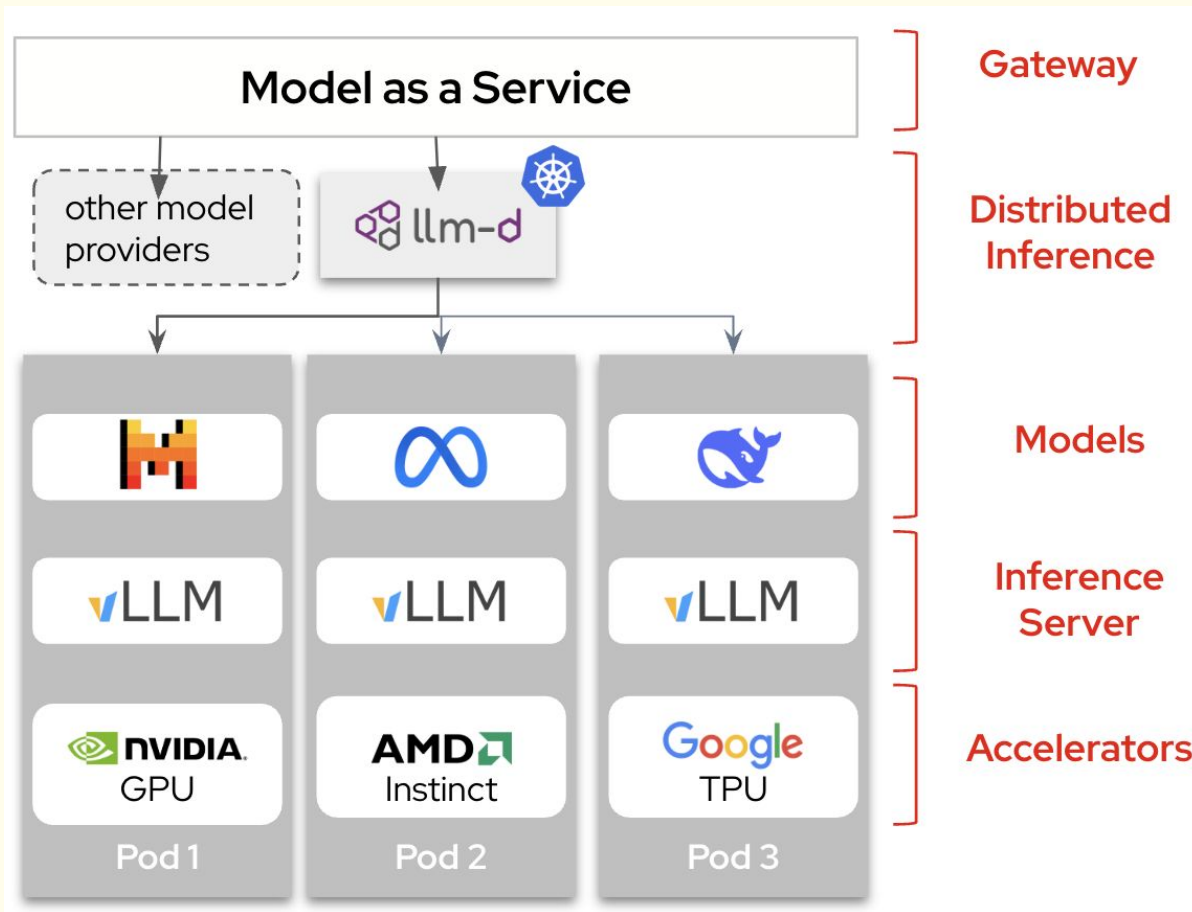


The bigger the model, it will:

- Be smarter
- Be slower
- Require more hardware

By compressing we are using our big model but not getting the downsides of it being slower or requiring more hardware

Machine for everyone



Machine for everyone|
!!! Demo !!!



**“Keep your
options open!”**

maarten-vandeperre.be

